Work Requirements and Child Tax Benefits*

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Abstract

Many U.S. safety-net programs condition benefit eligibility on work. Eliminating work requirements would better target benefits to the neediest families but would also attenuate pro-work incentives. We study how expanding child tax credits to non-workers affects maternal labor force participation, using administrative tax records and variation in state credit eligibility from quasi-random birth-timing. We employ a novel method for using placebo analyses to maximize the precision of our regression discontinuity estimator. We estimate a precise null effect from eliminating work requirements on mother's labor force participation; our 95% confidence interval excludes effects over one-third of one percentage point.

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1 Introduction

A large and growing body of research documents substantial long-term benefits of transferring resources to children growing up in poverty.[1] At the same time, many social welfare programs targeted at children exclude the lowest-income families from their scope. In the United States, for example, the federal Earned Income Tax Credit (EITC), the federal Child Tax Credit (CTC), and most state-level tax credits for children provide no assistance to parents without income and provide only limited assistance to parents with very little income. Relaxing these work requirements would increase the benefits received by the neediest children. However, when child tax credits condition eligibility on work, they create an extra incentive for parents to work. Removing work requirements from child tax credits thus risks attenuating the pro-work incentives that these programs create, along with any corresponding benefits of parental work on child outcomes. If parents can access child tax benefits whether or not they work, how will that affect their decisions about whether to participate in the labor force?

To shed light on this question, we study state variation in the design of child tax benefits. A number of states have recently adopted or expanded tax credits for taxpayers with children, and many more are currently considering such policies.[2] These credits are generally targeted at low- to middle-income families, with annual benefits ranging from $500 to $1,200 per child—comparable in magnitude to changes in benefit generosity frequently studied in the CTC and EITC literatures.[3] In this paper, we focus on six states that recently created tax credits for children below age six—an age group for whom the benefits of additional

[1] Recent studies include Dahl and Lochner (2012); Aizer et al. (2016); Bastian and Michelmore (2018); Cole (2021); Barr, Eggleston and Smith (2022); Bailey et al. (2024); Bhardwaj (2023); and Rittenhouse (2025). For reviews, see National Academies of Sciences (2019) and Aizer, Hoynes and Lleras-Muney (2022).

[2] Many if not all of these state policies were in response to failed federal legislation that would have permanently expanded the federal CTC. Concerns that this legislation would lead to parents dropping out of the labor force were central to its opposition (e.g. Kessler, 2021).

[3] For example, the largest expansions in EITC generosity involve total benefit increases of $1,700 to $1,900 in 2019 dollars (Meyer and Rosenbaum, 2001; Bastian, 2020). For comparison, the maximum per child federal CTC benefit is $2,000 today and was $1,000 prior to 2018. As detailed below, many of the state child tax credits we study phase out at lower income ranges than their federal counterparts.

resources are likely to be substantial (Bailey et al., 2024).[4] Importantly for our empirical design, these credits vary from one another in whether they condition benefit eligibility on work: two states (California and Colorado) introduced child tax credits that, like the federal CTC, were available only to taxpayers with positive earned income, while four states (Maryland, New Jersey, Oregon, and Vermont) introduced credits that did not include a work requirement.[5] In addition, midway through our sample period, California reformed its credit to eliminate its work requirement.

We study how the design of these child tax credits affects maternal labor force participation.[6] To do so, we consider variation in mothers' exposure to the policies induced by quasi-random variation in the timing of their children's births. In each state, children who turn six just after the end of the year are eligible for the credits whereas those who turn six just prior to the end of the year are not. We link this variation to administrative tax records to estimate the difference in labor force participation between mothers of age-eligible children versus age-ineligible children. We focus in particular on the elimination of California's work requirement, which allows us to directly isolate the effect of the work requirement on labor force participation in that state.

To increase the precision of our estimates, we propose and implement a novel placebo-based tuning analysis to guide the choice of specification for our regression discontinuity (RD) design. The procedure selects the specification that minimizes the mean-squared error of estimated pseudo-treatment effects during a set of placebo years prior to the initial adoption of the policy. We use this approach to optimize the RD bandwidth and polynomial order, along with other specification choices.

[4]A number of other states provide tax credits for parents of older children. A challenge with studying those credits empirically is that many have eligibility thresholds that coincide with eligibility for other tax credits—e.g., the federal CTC in the case of Arizona, Idaho, Maine, New York, and Oklahoma; the Child and Dependent Care Credit in the case of Massachusetts; and the EITC in the case of New Mexico.

[5]Although both California and Colorado conditioned eligibility on positive earnings, the design of their respective work requirements differed. Colorado's work requirement mirrored that of the federal CTC, with the credit gradually phasing in with income. In contrast, California provided the full credit amount to taxpayers with any positive earned income, regardless of how little.

[6]We focus on mothers because there is substantially more consensus regarding the magnitude of labor supply elasticities for working age males, which are thought to be quite small (e.g., Chetty et al., 2011).

We begin by estimating the effect of the child tax credits on the labor force participation of previously working mothers—the group whose potential exit from the labor force has been at the center of debates surrounding the federal CTC. Beginning with California, we estimate the effect of credit eligibility on labor force participation separately for the time period in which the credit was subject to a work requirement and separately for the time period in which it was not. The difference in these estimated effects yields our estimate for the effect of eliminating the work requirement. The estimate is a precise null, with a 95% confidence interval excluding effect sizes larger than 0.28 percentage points relative to a baseline participation rate of 93%. We similarly find a precise null effect on labor force participation in Colorado, the other state with a credit that conditioned eligibility on work. There, the 95% confidence interval ranges from -0.33 to 0.39 percentage points, again relative to a baseline of 93% participation. Finally, turning to the four states without work requirements, we find no evidence that any state's credit substantially affects labor force participation; averaging the estimates for the four states yields a 95% confidence interval of -0.28 to 0.17 percentage points (relative to baseline participation rates of 93 to 95%).

We next translate our estimates into labor participation elasticities. Doing so requires disentangling the income and substitution effects underlying our estimates. For a mother whose earnings, should she choose to work, would be below the relevant phase-out threshold, a credit with a work requirement would raise the financial return to working (a positive substitution effect). In contrast, a credit that was not conditioned on work would generate a pure income effect, in that her after-tax income would be higher whether or not she works. For a mother whose earnings would be above the credit's phase-out threshold, a credit that is not conditioned on work would reduce the after-tax benefit of working, thus generating a *negative* substitution effect. We apply this logic to convert our estimates into extensive-margin substitution and income elasticities, drawing on the variation in credit design and in taxpayers' incomes relative to the credit phase-out schedule for their state. Doing so yields an average extensive-margin substitution elasticity of 0.047 and a near-zero average

extensive-margin income elasticity.

A limitation of the administrative tax data on which we rely is that we cannot observe the state of residence for mothers who did not file a tax return and who are not listed on third-party information return.[7] Our main analysis circumvents this concern by focusing on exits from the labor force among previously working mothers — a group for whom an address is generally available via a prior-year information return. To expand our analysis to study flows *into* the labor force, we consider two additional samples: mothers whose children were enrolled in Medicaid and mothers of all children assigned a social security number constructed based on the Social Security Administration's Numident file. Using these samples, we find some evidence that credit eligibility may modestly increase labor force entry among previously non-working mothers; however, these effects are equally present for tax credits with and without a work requirement, suggesting that they may not be attributable to the work requirement itself.

Our results contribute to a growing literature that studies the labor supply effects of work requirements in U.S. safety net programs, such as SNAP, Medicaid, and TANF.[8] There are a number of important differences between the work requirements that govern these programs and those that govern the tax credits we study. For example, SNAP participants must work (or participate in job training) at least 80 hours per month, whereas at least a portion of most tax credits is available to taxpayers with even minimal positive earnings. At the same time, depending on one's wage, many more hours of work may be required to qualify for

[7]Tax records are available for a large share of U.S. adult residents. Auten and Gee (2009) estimate 86% of U.S. adults appear on a filed tax return. In a supplemental analysis, we estimate that approximately 96% of children born in 2016 or 2017 and issued a social security number appear on a tax year 2022 tax return (including a third party information return) that can be linked to one of the 50 states or D.C. However, individuals who cannot be linked to a tax return are primarily those who do not participate in the labor force (or for whom one's parent does not participate in the labor force) – the outcome on which we are focused. For discussion of these issues, see Larrimore, Mortenson and Splinter (2016); Gee et al. (2026).

[8]Several of these studies find no evidence of substantial employment effects from work requirements (Sommers et al., 2020; Han, 2022; Gray et al., 2023; Cook and East, 2026), whereas others find that conditioning benefit receipt on work leads to increases in employment (Harris, 2021; Falk, 2023). An older literature studies the elimination of the 1990s welfare reform, which involved changes in benefits for non-workers as well as reductions in effective marginal tax rates due to benefit phase-outs (e.g., Moffitt, 1992; Hoynes, 1997; Ziliak, 2015).

a tax credit's maximum amount.[9] In addition, many U.S. transfer programs require high levels of enforcement and verification multiple times a year; in contrast, participation in tax credits is based on self-certification of annual earnings with enforcement through ex-post audits. Because of these differences, the labor supply responses to tax and non-tax work requirements may be quite different.

Closer to our focus, several recent papers study the labor supply effect of a temporary reform that eliminated the federal CTC's work requirement and phase-in structure for tax year 2021 (Lourie et al., 2026; Enriquez, Jones and Tedeschi, 2023; Schanzenbach and Strain, 2023; Pac and Berger, 2024; Ananat et al., 2024). Consistent with our results, these studies generally find no evidence that the one-year suspension of the federal CTC work requirement affected labor force participation.[10] We complement these studies—which largely rely on survey data and an event-study design using differences in CTC eligibility (e.g., parents versus non-parents) in time periods with and without the CTC expansion—in two important ways. First, we add precision through a much larger administrative data set. Second, we employ an identification strategy that takes advantage of a particularly close link between the treated and untreated groups and that avoids concerns of differential labor market trends among parents of young children in the years surrounding the COVID-19 pandemic.

Apart from studies of the 2021 federal CTC reform, direct evidence on the effects of child tax credit work requirements has not been available.[11] As a result, efforts to predict the effect of various child tax credit reforms have largely relied on empirical estimates drawn

[9]To illustrate, in 2023 a single taxpayer with one child must earn $24,800 to reach the maximum federal CTC amount, corresponding to 40 hours of work at an hourly wage equal to 1.6 times the federal minimum wage.

[10]This literature also explores labor participation effects for various subgroups and largely finds similar null effects. One exception is Schanzenbach and Strain (2023), which reports a negative employment effect for the subgroup comprised of unmarried, low-income mothers with low education who have young children.

[11]Several prior papers study other labor supply incentive effects of the federal CTC, also using quasi-random variation in birth timing (Feldman, Katuscak and Kawano, 2016; Wingender and LaLumia, 2017; Mortenson et al., 2018; Lippold and Luczywek, 2024). Because these papers rely on variation in benefit amount for workers or in the timing of benefit receipt, as opposed to variation in the after-tax return to work, they do not speak to the effects of conditioning benefit eligibility on work. There also exists a large literature on the labor supply effects of child benefits outside the U.S. (Milligan and Stabile, 2007; González, 2013; Schirle, 2015; Messacar, 2021; Baker, Messacar and Stabile, 2023; Jensen and Blundell, 2024); these policies operate against a different institutional backdrop and set of preexisting labor participation incentives.

from policy variation in the EITC (Corinth et al., 2021; Goldin, Maag and Michelmore, 2022; Bastian, 2024).[12] However, there are several reasons to think that the state child tax credits we study provide a better guide for understanding the effect of reforms that would relax the work requirements governing state or federal child tax credits. First, child tax credits operate in the shadow of the EITC and the sizable labor supply incentives that it generates; it may be that adding or subtracting incentives on top of those generated by the EITC does not much affect most parents' decisions about whether to work. Second, policies like the EITC incentivize work by providing a benefit that is exclusively available to working taxpayers. In contrast, reforms that delink child benefits and work (like the ones we study) affect the return to work by expanding an existing benefit to non-workers. Although both types of policies shape the return to work, only the latter does so for taxpayers with incomes above the EITC phase-out; in some analyses, a substantial share of the predicted reduction in labor participation from expanded child tax benefits stems from the labor response of this group (e.g., Corinth et al., 2021). Third, many of the EITC elasticity estimates are estimated from policy reforms that took place in the 1990s or earlier; yet the nature of female labor force participation may be quite different today (e.g., Blau and Kahn, 2007), and indeed more recent EITC reforms have yielded smaller estimated labor supply responses (Schanzenbach and Strain, 2021; Kleven, 2024). We therefore view our results as providing some of the most credible evidence to date on the effects of conditioning child tax credits on work.

A final contribution of our paper is methodological. Researchers exploiting quasi-random treatment assignment in regression discontinuity settings must make a number of modeling choices relating to the empirical specification they employ (Lee and Lemieux, 2010; Cattaneo, Idrobo and Titiunik, 2019). Popular existing methods to select RD specifications aim to minimize estimated first-order asymptotic approximations to the mean squared er-

[12]Much of the EITC literature on which these studies draw finds large labor force participation effects of the credit (for reviews, see Eissa and Hoynes, 2006; Nichols and Rothstein, 2016; Schanzenbach and Strain, 2021), though two recent reappraisals provide evidence against this conclusion (Kleven, 2024; Looney, 2024). In addition, newer studies that do find positive labor supply effects tend to find smaller elasticities than those reported in the earlier literature (e.g. Chetty, 2012; Hoynes and Patel, 2018; Bastian and Jones, 2021).

ror (MSE) (Imbens and Kalyanaraman, 2012; Calonico, Cattaneo and Titiunik, 2014; Pei et al., 2022). In contrast, our procedure evaluates candidate specifications using their out-of-sample performance in placebo settings that preserve the actual birth date cutoff used for identification in the target setting of interest. When the available placebos approximate the relevant bias-variance trade-offs for the target setting, this method allows us to choose among potential specifications to minimize MSE while avoiding concerns of over-fitting or excessive regularization.[13] We provide evidence that our method yields substantial performance gains in settings like the one we study in which suitable placebo datasets are available to the researcher.

2 Institutional Background

The U.S. income tax code provides a number of benefits for taxpayers with children. As of 2023, the federal Child Tax Credit (CTC) provides a tax credit of up to $2,000 per child under the age of 17, and the full credit amount is available for taxpayers with incomes up to $200,000 if single or $400,000 if married.[14] Taxpayers without income receive no benefit from the CTC, and many low-income taxpayers do not qualify for the full credit amount (Collyer, Wimer and Harris, 2019; Goldin and Michelmore, 2022). The other main federal tax credit primarily aimed at taxpayers with children is the Earned Income Tax Credit (EITC). The EITC phases in by income, with a maximum benefit in 2023 ranging between $3,995 and $7,430 depending on the number of children a taxpayer claims.[15]

Turning from federal to state policy, a number of states provide their own child tax

[13] Ludwig and Miller (2007), Imbens and Lemieux (2008), and Kettlewell and Siminski (2026) propose related approaches that evaluate RD specifications out-of-sample based on their performance recovering placebo pseudo-effects. Whereas tese approaches measure performance at other values of the running variable (away from the true RD cutoff), our approach measures performance at the same birth date cutoff that determines credit eligibility, but in placebo pre-period years prior to the credit's actual adoption.

[14] The CTC has been reformed a number of times since its introduction in 1997; we focus on the rules in place for recent tax years other than 2021. For tax year 2021, the federal CTC was available to taxpayers without income from work and the maximum benefit was increased to $3,600 per child younger than age 6 and $3,000 per child older than six and younger than 18.

[15] A smaller EITC benefit is available to working taxpayers who do not claim children. For most children, the maximum age to qualify a taxpayer for the EITC is 18, or 23 if the child is a full-time student.

credits.[16] The details of these credits vary across states by benefit size, whether the credit is refundable, the age restrictions for eligible children, and income restrictions (including work requirements) on the taxpayer claiming a child. We focus on state tax credits in six states—California, Colorado, Maryland, New Jersey, Oregon, and Vermont—that provide benefits to parents of young children. To qualify for the credit in these states, taxpayers must claim at least one child under age six, where the child's age is evaluated as of December 31st of the given tax year. Appendix Table A.1 reports the average benefit amounts available through these state and federal tax credits for low-income mothers with young children in the states on which we focus.

In 2019, California became the first state to offer a tax credit with this design. When first introduced, the Young Child Tax Credit (YCTC) provided a maximum credit of $1,000 per tax return for taxpayers who claimed a child below the age of six.[17] The credit targets low-income families and gradually phases out between $25,000 and $30,000 of income. From 2019 through 2021, the credit had a work requirement: claimants were required to have positive earned income. In these years, taxpayers without earned income did not qualify for the credit whereas taxpayers with $1 or more of earned income qualified for the full benefit amount, assuming they were otherwise eligible (Figure 1, Panel A).[18] California then eliminated the credit's work requirement starting in 2022. After the reform, taxpayers without income could qualify for the full credit amount (Figure 1, Panel B).

In 2022, Colorado introduced the Colorado Child Tax Credit. Like California's credit, the Colorado credit was targeted at lower-income families: the credit was fully phased out for single parents earning over $75,000 and married parents earning over $85,000. Colorado's credit was somewhat more generous than California's in two respects. First, it had a larger maximum benefit for a taxpayer claiming one child ($1,200). Second, Colorado's credit

[16]Many states also provide state-level EITCs, which are typically provided as a fixed percentage of the federal EITC (the California EITC is an exception). See Appendix Figure A.1.

[17]This amount was increased to $1,083 and $1,117 for tax years 2022 and 2023, respectively.

[18]During this time period, California expanded eligibility for taxpayers without social security numbers authorizing them to work; below, we consider specifications that exclude this group from our analysis.

was provided on a per-child basis (e.g., taxpayers with two children could receive up to $2,400) whereas California's maximum benefit applied per-return. Like the initial design of California's credit, the Colorado credit had a work requirement during our sample period. However, unlike in California, the Colorado credit was pegged to the federal CTC so that its benefits phased in gradually with earned income (see Figure 1, Panel A).[19]

Finally, during 2022 and 2023, four states adopted child tax credits for parents of young children that were not conditioned on work: Maryland (2023), New Jersey (2022), Oregon (2023), and Vermont (2022). Like the federal CTC, these state credits were each provided on a per-child basis, with maximum credit amounts of $500 per child (Maryland, New Jersey in 2022) or $1,000 per child (Oregon, Vermont, New Jersey in 2023). Although all four credits were available to non-workers, the credits varied widely with respect to the maximum income at which taxpayers could qualify ($15,000 for Maryland to $175,000 for Vermont) as well as the phase-out rate (steeper in Maryland and Oregon, more gradual in New Jersey and Vermont). Panel B of Figure 1 provides additional detail.

It is worth noting that the benefit variation we study is comparable to the magnitude of benefit variation frequently studied in the literature on tax credits and extensive-margin labor supply. For example, Feldman, Katuscak and Kawano (2016) and Lippold (2022) study the effects of a child aging out of federal CTC eligibility during a time period in which this resulted in a maximum benefit change of $1,000. With respect to the EITC, studies using variation induced by the 1986 Tax Reform (Eissa and Liebman, 1996) or changes in state EITC policy (Michelmore and Pilkauskas, 2021) focus on benefit changes that are smaller than the variation we study (roughly $300 to $700 in 2019 dollars), whereas studies focusing on the EITC's introduction (Bastian, 2020) or 1993 expansion (Meyer and Rosenbaum, 2001) involve larger benefit changes.[20] Appendix Figure A.2 plots the percentage change in the

[19]Because the Colorado credit is based on the Federal CTC, it reflects the latter's limits on refundability—hence the two-tiered effective phase-in regions shown in Figure 1.

[20]The federal EITC's introduction involved a benefit change of $1,900 in 2019 dollars. Studies of the Omnibus Budget Reconciliation Act of 1993 typically compare benefit changes of taxpayers with one versus two or more children, corresponding to up to approximately $1,700 in 2019 dollars.

after-tax return to work from these reforms along with the change from the elimination of California's work requirement.[21] Because of the lack of a phase-in, the elimination of the work requirement led to a very large reduction in the return to work for the lowest-income working mothers, with a smaller (and monotonically declining) reduction for middle- and higher-income mothers. In contrast, the EITC reforms generated a smaller percentage change in the return to work for the lowest-income taxpayers—who did not qualify for the full EITC amount—and a larger percentage change for middle- and higher-income taxpayers.

3 Data

We draw on federal administrative tax records for our analysis. This data includes the universe of children who receive a social security number as well as the mothers listed on those children's birth certificates.[22]

Our main sample consists of mothers who worked during the prior year and whose focal child turns six within a window of dates surrounding the turn of the calendar year. We construct this sample as follows. The first three steps each draw on Social Security birth records. First, we identify the cohort of children for a given outcome year (i.e., the year for which we will measure maternal labor supply). This cohort consists of the universe of U.S. children who turned six years old during the final months of the outcome year as well as those children who turned six during the first months of the year following the outcome year. For example, the 2020 cohort consists of children born at the end of 2014 and the start of 2015. Children in the latter category are age-eligible for the credit in tax year 2014 whereas children in the former category are not. Second, we link children to the individual whose Social Security number is listed as the child's mother on the child's birth certificate. Third, for our California analysis, we identify other children of the same mother, and drop

[21] For additional context, Appendix Figure A.3 provides information about the distribution of earnings for mothers with young children.

[22] A high fraction of birth certificates contain maternal information (93.5%); by comparison, 79.9% contain paternal information.

mothers who have given birth to a child younger than the focal child before the outcome year. This restriction excludes mothers who would continue to qualify for the credit despite their reference child aging out of eligibility. We do not apply the same restriction in others states as those state credits are available on a per-child basis. Fourth, we restrict the sample to the subset of mothers who received a third-party information return (Form W-2, 1099-Misc, or 1099-NEC) showing positive income for the year prior to the outcome year.[23] We assign individuals to states based on the taxpayer's residence information listed on the applicable information return. Finally, our main analyses exclude the cohort of mothers whose children turn six around the end of 2021 because the temporary expansion of the federal CTC for that year provided additional benefits for taxpayer claiming children under age 6—the same eligibility cutoff we use for identification.

We supplement our analysis with two alternative samples that include non-working mothers. Recall that our main sample addresses the challenge of assigning mothers to states by requiring a prior-year information return (which includes the mother's address). Our alternative samples take different approaches to the data challenge that arises when this requirement is dropped. First, we construct samples of mothers whose children were enrolled in Medicaid or Children's Health Insurance Program (CHIP) during the prior year. Crucially for our purposes, the children in these categories are listed on Form 1095-B, showing their health insurance coverage, which allows us to identify their state of residence even if their mother did not receive third-party reported income during the prior year. Our second alternative sample consists of the universe of children who were born in California. This sample is constructed based on the Social Security Administration's Numident file, which we accessed and linked to tax records through the U.S. Census Bureau.[24] Each of these alternative samples has its own benefits and limitations. The Medicaid sample has good coverage of low-income mothers

[23]We do not otherwise limit our main sample based on income because eliminating the work requirement affects the return-to-work for individuals with incomes above the credit eligibility thresholds. We present subgroup results for more granular income groups.

[24]The Numident file contains dates of birth, county of residence at time of enumeration, and basic demographic information. We link children to parents using the probabilistic matches contained in the Census Household Composition Key; see Aldana (2022) and Bernard, Drotning and Genadek (2024).

but may not be representative of the uninsured or individuals who obtain health insurance through other means. In contrast, the Census sample has broader coverage, but lacks the information return data for self-employment income that our primary data include.[25] Across the states we consider, the size of the Medicaid sample ranges from 58 to 80% of the size of our main sample. The Census sample size is roughly twice our main California sample.

Our primary outcome is whether an individual works during the outcome year. It is constructed based on whether the IRS receives an information return (Form W-2, 1099-Misc, or 1099-NEC) for the individual for the year showing positive income. Because this measure is based on third-party filed information returns, it covers individuals who did not themselves file an income tax return. Using information returns increases the likelihood that our estimates measure real changes in labor force participation as opposed to changes in reporting behavior (Garin, Jackson and Koustas, 2025). In supplemental analyses, we also consider labor force participation measured using the income reported on individuals' filed tax returns, which is available to us for tax years 2022 and earlier.[26]

4 Empirical Framework

4.1 Identification Strategy

Our empirical strategy relies on variation in credit eligibility induced by quasi-random variation in the date that a child is born. Specifically, we compare labor force participation of mothers whose child turns six before the end of a given year with mothers whose child turns six after the start of the subsequent year. We make these comparisons using regression

[25] In addition, there is a substantial time delay between when most children receive a Social Security Number and their sixth birthday, which could lead to inaccurate assignments for children who move into or out of the state in the intervening years.

[26] A limitation of all our outcome measures is that they do not capture informal work arrangements that go unreported by both the taxpayer and employer; hence, an apparent labor supply response may actually reflect a shift from the informal to the formal sector.

discontinuity specifications of the following form:

$$Y_{it} = \alpha + \beta \, \mathbf{1}_{\{DOB_i 0\}} + g_1(DOB_i) + g_2(DOB_i) \, \mathbf{1}_{\{DOB_i 0\}} + \gamma_t + \varepsilon_{it} \tag{1}$$

where Y_{it} is an indicator for whether mother i had positive earned income in tax year t; DOB_i indicates the date of birth of the focal child, centered around the turn of the year with December 31 of year t denoted by 0; γ_t is a set of year fixed effects; and $g_1(\cdot)$ and $g_2(\cdot)$ are local polynomials. In this specification, the effect of having a child of an age that qualifies a mother for the credit is given by β.

To interpret β as the causal effect of age-eligibility for the credit, we impose the standard RD identifying assumption that, but for the policy, maternal labor force participation is continuous in birth date across the age-eligibility cutoff. This assumption would be violated if the mothers of children born on either side of the age cutoff differed discontinuously in their propensity to work for reasons unrelated to the state CTC, such as from compositional differences in the timing of births or if other relevant policies vary based on the same age cutoff–such as school entry rules or eligibility for other benefits. We are unaware of any other policies affecting the mothers in our sample that rely on this age cutoff, other than the 2021 federal CTC expansion described above; e.g., kindergarten entry occurs in the year *prior* to aging out of the credit. In practice, because finite sample sizes require us to estimate the cutoff effect using observations away from the threshold, the validity of a particular estimator additionally depends on whether its functional form accurately reflects the relationship between potential labor force participation and child birth date over the estimation window. When these conditions hold, β captures the causal effect of credit eligibility for mothers whose child's age is near the threshold for credit eligibility.

Because our main sample is limited to mothers who worked during the prior year, our RD estimand primarily represents exits from the labor force by previously working mothers in the year that their child ages out of credit eligibility.[27] Intuitively, for a work-contingent

[27]Note that if a credit shapes mothers' labor participation decisions, the composition of this sample in a

credit, this group would include mothers who would exit the labor force were it not for the credit but who are induced by the credit's incentives to remain in the labor force for their child's one additional year of eligibility. In our alternative samples that include mothers who did not work in the prior year, β is additionally identified by previously non-working mothers whose decisions about whether to enter the labor force may be affected by whether their child makes them potentially eligible for the credit (see Section 5.5).

Finally, for our analysis of the California reform, we estimate the effect of eliminating the work requirement by comparing the estimated values of β across policy periods (i.e., for the work requirement years versus the years without a work requirement). To interpret this difference as the causal effect of eliminating the work requirement requires an additional assumption: but for the elimination of the work requirement, the effect of the credit on labor force participation would have been the same in both policy periods. Below, we provide evidence for the plausibility of our identifying assumptions for our setting.

4.2 Placebo-Based Tuning Analysis

In this subsection, we propose and implement a data-driven method for selecting the empirical specification in (1). The method selects between specifications based on performance in placebo pre-periods; we therefore refer to it as placebo-based tuning (PBT). We use this approach to determine: (1) bandwidth (the width of the birth-timing window to include); (2) order of the local polynomial in children's dates of birth (linear, quadratic, or levels); (3) kernel (uniform, triangular, Epanechnikov); and (4) whether to exclude children born around the end-of-the-year holidays through a "donut" specification (e.g., Barreca et al., 2011).

To make these determinations, we evaluate the performance of each candidate specifica-

particular year will partly reflect the effect of the credit in any prior years in which the credit was in place. That is, if a credit has been in effect for multiple years, some fraction of the previously working mothers may only have been working because they qualified for the credit. In this case, the estimated effect of the credit would reflect such mothers exiting the labor force when their child ages out of credit eligibility or when the work-conditional credit is replaced by an unconditional credit (as in California). In contrast, when we expand our sample to include non-working mothers, the subgroup of mothers that identifies our treatment effect is no longer dependent on the effect of the credit from prior years.

tion applied to estimating the effect of placebo policies in the same state and at the same age-cutoff during the years before the credit took effect. Specifically, we separately estimate the effect of each pseudo-policy in each year before the credit is adopted, treating the turn-of-the-year as the threshold for eligibility. We evaluate performance based on the mean of the squared errors (MSE) from the estimated effects of the pseudo-policies. Because the true "effect" of each pseudo-policy is zero, the squared error for a pseudo-policy corresponds to the square of its estimated treatment effect.[28] We then choose among the alternative specifications to minimize the MSE from this analysis.

We first consider the optimal bandwidth and order of the local polynomials. On the one hand, narrower birth-timing windows could reduce bias by estimating our effects from groups that are more similar to one another. Similarly, it could be that bias is reduced by flexibly adjusting for differences in children's birth dates. On the other hand, narrower birth-timing windows and more flexible functional forms could reduce the precision of our estimates by yielding an estimator with larger variance.

We empirically evaluate these trade-offs in Figure 2 for each of the six states. Consider the specification for our California analyses (Panel A). The blue line plots RMSE for specifications that do not adjust for differences in birth dates between groups – i.e., for simple comparisons of means. For specifications in this category, RMSE follows a "U"-shape pattern with respect to birth-window width, consistent with the presence of a bias-variance trade-off. In particular, RMSE declines monotonically for narrow birth-windows, with larger samples reducing the variance of the estimator from sampling uncertainty. For wider birth-windows, RMSE is monotonically increasing, consistent with increasing bias as the groups of mothers being compared becomes less comparable. The RMSE of the estimator is minimized at a 4-month birth-window. Interestingly, this width corresponds to the widest birth-window that does not intersect with California's kindergarten cutoff of September 1.

[28]We begin this analysis in 2005 rather than 2000 (which is the first year for which we have outcome data) due to differences in the availability of W-2 data prior to that year, which could shape the bias-variance trade-off of candidate specifications.

The yellow and purple lines in Figure 2 respectively correspond to specifications that employ local linear or local quadratic polynomials. For California, for specifications that use birth-windows of 8 months or less, the linear polynomial yields higher RMSE than the simple comparison of means, and the quadratic polynomial yields higher RMSE than both. For windows that include a year or more on either side of the cutoff, both the linear and quadratic specifications yield a lower RMSE than the levels comparison. Intuitively, wider windows involve comparing mothers who may differ substantially in their composition, increasing the importance of adjusting for differences in the dates of their children's births. Across the combinations we consider, the level estimator with a four-month window yields the lowest MSE.[29]

We next consider additional specification choices relating to the type of kernel employed for local polynomial estimation and of the exclusion from our sample of children born around the turn-of-the-year (a "donut" specification). We find that the relative performance of alternative kernels is closely linked to the local polynomial order and bandwidth (Appendix Figure A.4). Based on these results, we opt for a uniform kernel, which has the additional virtue of being the most straightforward to implement and interpret. Donut specifications appear to weakly reduce estimator precision, especially for narrower bandwidths (Appendix Figure A.5), so we do not employ one.

Based on these results, our preferred specification for California is a simple difference in means among mothers whose child turns six within four months of the turn-of-the-year:

$$Y_{it} = \alpha + \beta \, \mathbf{1}_{\{DOB_i 0\}} + \gamma_t + \varepsilon_{it}. \tag{2}$$

[29] An assumption underlying the PBT approach is that the ranking of candidate estimators' performance is similar across the placebo and policy periods; this could be violated by temporal differences in sample size, outcome variance, the distribution of birth dates, or the shape of the potential untreated outcome function, in which case the placebo results could present a misleading picture of the bias-variance trade-off in the policy period. Although we cannot test this assumption directly, we can assess whether the relative performance of candidate specifications is stable across different portions of the pre-period. We find a broadly similar pattern of specification performance using the first versus second halves of the pre-period as our placebo (Appendix Figure A.6).

Conceptually, the PBT results for California suggest that allowing for birth date trends does not improve estimation accuracy enough to offset the resulting increase in variance.

For each of the other five states, we apply the same PBT procedure to select the MSE-minimizing local polynomial order and bandwidth (Figure 2, Panels B through F). These state MSE plots share many of the same qualitative features, with the levels comparison (i.e., the simple difference in means) yielding the lowest MSE for narrow bandwidths while the local linear specification yields lower MSE for wider bandwidths. Interestingly, the local quadratic specification appears dominated at each bandwidth we consider. For Colorado and Vermont (like California), the minimum MSE is obtained from levels comparisons with moderate bandwidths (5 to 9 months), whereas for Maryland, New Jersey, and Oregon, linear specifications with very wide bandwidths (19 to 24 months) yield slightly lower MSE than the optimal levels specification. Below, we primarily rely on the MSE-minimizing specification for each state, while also considering alternative specifications with comparable MSE.

Our final set of analyses in this section compares the performance of our PBT analysis to alternative approaches for selecting RD specifications. One natural comparison is the specification employed in a number of recent high-quality papers that employ a similar identification strategy to ours, exploiting quasi-random variation in birth timing to assess the effect of eligibility of some program (including Barr, Eggleston and Smith, 2022; Lippold and Luczywek, 2024; Garin, Jackson and Koustas, 2025). Those papers primarily rely on a local linear polynomial with a one-month bandwidth and a donut-design that excludes children born in the narrow window around the turn-of-the-year. We also compare PBT to the popular Calonico, Cattaneo and Titiunik (2014) ("CCT") bandwidth selection method.

We evaluate PBT against these alternatives using the states that did not adopt a child tax credit for children under age 6 during our sample period. For each state, we determine the optimal specification based on minimizing the RMSE from 2005-2018, and then use that specification to estimate the pseudo-effects of a placebo credit during each of our main outcome years (2019, 2020, 2022, and 2023). Appendix Figure A.7 presents the results of

this comparison. The distribution of estimates from each method is centered at zero, but PBT specifications are substantially more precise than either alternative.

5 Results

5.1 California Results

We present summary statistics for our primary sample of recently working California mothers in Appendix Table A.2. The large majority (93%) of mothers in our sample filed a prior-year tax return; of this group, approximately one-third claimed the federal EITC and 90% claimed the federal CTC. Their average (individual) income was approximately $57,000.

We observe several differences between mothers in our treatment and control groups. As expected, mothers of children who turn six at the end of the year tend to be slightly older than mothers of children who turn six at the start of the subsequent year. In addition, we observe statistically significant but small (< 1 percentage point) differences in tax filing, and among federal tax filers, marital status and claiming of the EITC and CTC.

Figure 3 reports labor force participation for California mothers by child birth date, separately for outcome years in which credit eligibility was conditioned on work (Panel A) and for outcomes years in which it was not (Panel B).[30] The red lines denote mean labor force participation for children born in a 4-month window around the turn-of-the-year—the PBT-selected specification—estimated separately for mothers on either side of the cutoff.

Table 1 reports the corresponding regression results. Column 1 reports the estimated effect of the California credit during the period in which a work requirement was in place. We find that mothers of age-eligible children are 0.11 percentage points (11 basis points)

[30]Mean labor force participation rates by year and treatment group are reported in Appendix Figure A.8. In all but one year (2017), the groups are not statistically significantly different from one another. We are not aware of a policy cause for the observed difference in 2017, and a joint test of the yearly differences does not reject the null hypothesis that the eligibility groups had equal labor force participation in each pre-period year. Additional analyses described in Section 5.3 similarly find near-zero effects when the multiple pre-period years are pooled together.

less likely to work, with a 95% confidence interval ranging from -0.32 to 0.09 percentage points. Column 2 reports the corresponding estimate for the time period following the work requirement's elimination. Here too we find that mothers of age-eligible children are 0.11 percentage points less likely to work (95% CI from -0.31 to 0.08). Subtracting the estimate in Column 1 from the estimate in Column 2 yields our estimate for the effect eliminating the credit's work requirement on labor force participation. The estimate corresponds to δ in the following specification:

$$Y_{it} = \alpha + \beta\, AgeEligible_i + \delta\, AgeEligible_i * Post_t + \gamma_t + \varepsilon_{it} \tag{3}$$

where *Post* indicates a year following the elimination of the work requirement.

As shown in Column 3, we estimate that the removal of California's work requirement led to a reduction in mothers' labor force participation of 0.001 percentage points (0.1 basis points), with a 95% confidence interval ranging from -0.28 to 0.28 percentage points. To put this magnitude in perspective, of the approximately 1.05 million working mothers with a child under age 6 in California in 2024 (U.S. Census Bureau, 2024), our point estimate implies a reduction of only 11 mothers from the labor force, with a 95% confidence interval translating to approximately 3,000 mothers entering or exiting the labor force. Hence, the results in Table 1 provide evidence against the hypothesis that eliminating the work requirement caused a substantial share of working mothers to exit the labor force.[31]

One factor that may help explain why removing the work requirement did not lead to a substantial reduction in labor force participation is that California's credit was itself a new policy. Immediately after enactment, taxpayers may not have been aware of the credit and therefore failed to consider its work requirement when making their labor participation

[31] Appendix Table A.3 reports three robustness checks. Panel A limits our sample to the 98% of taxpayers with a valid Social Security Number to ensure that changes in eligibility for non-citizens do not bias our results. Panel B limits our sample to the 84% of mothers who claimed the child on their prior-year tax return. Third, recall that the work requirement was in effect in 2021, but the age-eligibility threshold was conflated by the federal CTC expansion. To incorporate that year into our analysis, Panel C uses mothers in states without child tax credits as a control group, i.e., a triple-difference specification. For each exercise, we obtain results similar to those from our baseline analysis.

decisions. We assess this possibility in two ways. First, although we lack direct evidence on taxpayers' knowledge of the credit, we can indirectly estimate take-up based on public reports from the California agency that administers the credit. Among federal tax filers (a group composing 93% of our sample), we estimate high rates of take-up of the California credit, rising from approximately 84% in 2019 to 97% by 2020 (Appendix Table A.4). Although claiming a tax credit is not a perfect proxy for awareness—guided tax preparation software may result in taxpayers claiming credits of which they are unaware—taxpayers may learn about a credit through the process of claiming it on their return or through their interactions with tax preparers (Chetty and Saez, 2013).[32]

Second, if the recency of the credit's adoption is an important contributor to the small effects it generates, we would expect to see larger effects of the work requirement over time as a growing number of taxpayers become aware of the credit and incorporate it into their decision-making (Chetty, Friedman and Saez, 2013; Garin, Jackson and Koustas, 2025).[33] Appendix Table A.5 investigates the possibility that the effects of the credit may change over time by estimating the effect of eligibility separately for each year in which the work requirement was in effect. To account for the possibility that some taxpayers only learned about the policy in its third year, we include 2021 in this analysis. However, because the federal CTC provided different benefits for the age-eligible versus age-ineligible mothers in our sample during 2021, for this analysis we also include mothers in other states as an additional control group. In all three years, we estimate a very small effect of age-eligibility on maternal labor force participation, with no gradient that would indicate increasing effects of the work requirement over time.

A distinctive feature of California's work requirement was that a taxpayer would qualify

[32]To the degree taxpayers are not aware of the credit or its rules, their labor participation responses may diverge from how they would have behaved under perfect optimization. For purposes of extrapolating our results to other child tax benefits, this aspect of the credits is arguably a feature, not a bug, as many other tax policies would also not be perfectly salient. For example, even the federal EITC's work incentives are not salient for many taxpayers (Chetty and Saez, 2013; Chetty, Friedman and Saez, 2013).

[33]A different factor that could contribute to a smaller effect in the credit's first year is if there are fixed costs to adjusting labor supply; parents might find it worthwhile to incur the cost to receive the credit only if their children have more than a single year of future credit eligibility.

for the full credit amount as long as they earned at least $1 during the tax year. A natural question to ask is how the effects of a policy with this design would compare to the effects of a policy where benefits were conditioned on a higher earnings threshold. In Appendix B, we show that under mild conditions, the effect of a work requirement on labor force participation is weakly decreasing in the applicable earnings threshold. Intuitively, individuals who are marginal labor force participants with respect to a higher-dollar threshold would also be marginal with respect to a lower-dollar threshold, but not necessarily vice-versa.[34] Consequently, we interpret the effect of eliminating the California credit's work requirement as an upper bound for the effects of alternative work requirements tied to higher earnings thresholds.

5.2 Other State Child Tax Credits

We now turn from California to other states that offer child tax credits for parents of young children. One of these states (Colorado) conditions credit eligibility on work, whereas four states (Maryland, New Jersey, Vermont, and Oregon) do not. For each state, we select the birth date window and empirical specification using our placebo-based tuning analysis (see Figure 2). Labor participation outcomes by birth date for these samples are shown in Figure 3, and sample characteristics are provided in Appendix Table A.6.

Table 2 reports the effect of eligibility for these five state credits on labor force participation. We begin with Colorado, which limited credit eligibility to working taxpayers. As with the California credit when a work requirement was in place, we find that eligibility for the Colorado credit did not affect maternal labor force participation; our point estimate corresponds to an increase of 0.03 percentage points with a 95% confidence interval of -0.33 to 0.39 percentage points (Column 1).[35]

[34]For example, the federal CTC was initially available only to taxpayers with earned income above $10,000; any taxpayer who chose to work because of that incentive would also presumably decide to work if the required earnings threshold was reduced to $5,000. But some of those taxpayers might not choose to work if the required earnings threshold was instead raised from $10,000 to $15,000.

[35]We do not observe an increase in the effect of Colorado's work requirement over time; see Appendix

Columns 2 through 5 of Table 2 report results for the four states that provided tax credits to parents of children under the age of six that were not conditioned upon work. In none of these states do we find evidence that the credit reduced labor force participation: the estimated reductions range from 0.05 to 0.54 percentage points, and none is statistically significant. Combining the estimates across these four states (via a population-weighted average) yields a 95% confidence interval of -0.28 to 0.17 percentage points.

For each of the states considered in this subsection, the sign of the estimated labor force participation is as expected – i.e., positive in Colorado, which has a work requirement, and negative in the four states without work requirements. However, the magnitude of the estimated effects is very small. Taken together, the estimates from the five state child tax credits considered in this subsection are consistent with our finding from California that eliminating a child tax credit work requirement does not cause a substantial number of mothers to exit the labor force.

5.3 Robustness and Placebo Checks

To assess the RD identification assumption, we undertake a range of placebo analyses. We first investigate differences in potential outcomes across the age-eligibility threshold in pre-period years before a state adopts its child tax credit. To avoid biasing these estimates toward zero, we modify our primary RD specification, which was tuned to yield a null estimate for these pre-period years. In particular, we estimate each placebo year's pseudo-treatment effect using a leave-one-out-procedure, in which each placebo year's specification is selected to minimize the MSE during the *other* placebo years. These estimates are tightly distributed around zero (Appendix Tables A.8 and A.9, Column 1). We find similar results with several other RD specifications, described in more detail below and not based on our PBT approach, for estimating the pseudo-effects during the placebo years (Columns 2-5).

We next consider two exercises based on alternative placebos. First, we consider mothers

Table A.7.

living in states that did not adopt a tax credit for children below the age of six, in the years covered by our actual analysis. We plot these pseudo-effect estimates in Appendix Figure A.9 and report the implied difference-in-discontinuities estimate associated with the California reform in Appendix Table A.10. Second, we take advantage of the fact that, in California, the credit provides a $1,000 credit per tax return, not per child. As such, mothers of children near the age cutoff who also have a younger sibling see no difference in their eligibility in the year that the older child turn six. Replicating our main analysis for these placebo groups does not provide evidence of differences in labor force participation across the age-eligibility threshold that are unrelated to eligibility for the credit.

We next turn to our difference-in-discontinuities analysis for California, which requires that the average effect of the work requirement on labor force participation be the same in both policy periods. This assumption would be violated, for example, if the labor force environment differed across time periods in ways that could exacerbate or mute the effects of the work requirement. Of particular concern in our setting is that our study period for our California analyses overlaps with the onset of the COVID-19 pandemic in 2020. To assess this issue, Appendix Table A.11 repeats the main difference-in-discontinuities specification from Table 1 excluding 2020 (the first year of the pandemic) or excluding 2020-2022 (the years for which the pandemic was most disruptive); these analyses yield nearly identical results as our main specification.[36] Along similar lines, we compare the effect of the work requirement across the two years of the policy period for which it was in effect—2019 versus 2020—and find no evidence that the effect of the policy varied across these years.[37] Third, we plot the distribution of difference-in-discontinuities estimates for the placebo states that did not adopt a child tax credit for children under the age of six. Appendix Figure A.10

[36]This analysis also provides evidence against an alternative explanation for why the reform did not produce a large reduction in labor force participation, namely that mothers would not want to quit their jobs for only one additional year of credit eligibility. However, because the age-eligibility cutoff is anticipated, mothers identifying the 2023 effect would have faced two potential years of eligibility under the reformed credit rather than just one.

[37]The onset of the COVID-19 pandemic is of less concern for our analysis of the tax credits from states other than California because those credits were enacted in 2022 and 2023, after the most disruptive period of the pandemic concluded.

shows that the distribution of these estimated pseudo-effects is centered at zero, consistent with an absence of differential trends in labor force participation between the age-eligible and age-ineligible groups during our sample period.

Next, we consider the sensitivity of our results to alternative RD specifications with similar performance according to our placebo-based tuning analysis. For California, recall that our preferred specification entails a simple comparison of means; however, Panel A of Figure 2 showed that a local linear specification with an 18-month bandwidth achieved nearly as low of a mean-squared error.[38] Despite substantial differences from our preferred specification (linear polynomial versus simple comparison of means; 18-month versus 4-month bandwidth), this alternative specification yields estimates that are very similar to our main results (see Appendix Table A.12). In contrast, the local linear specification employed in recent papers that exploit quasi-random variation in birth timing (e.g., Wingender and LaLumia, 2017; Barr, Eggleston and Smith, 2022; Garin, Jackson and Koustas, 2025; Rittenhouse, 2025; Bhardwaj, 2023; Lippold and Luczywek, 2024) involve much smaller bandwidths that, according to our placebo-based tuning analysis, tend to yield larger mean-squared errors in our setting. We find that the local linear specification with a 1-month bandwidth used by Barr, Eggleston and Smith (2022), for example, yields imprecise null estimates, with estimated confidence intervals almost an order of magnitude larger than those obtained by our preferred specifications (Appendix Table A.13, Panel A). Panel B of the same table shows that specifications based on CCT-selected bandwidths also yield null effects, with larger confidence intervals than our preferred specifications.

A potential concern with our PBT method for specification selection is that, by training it to minimize error during a placebo period in which the true effect is zero, we are inadvertently selecting an estimator that performs well at obtaining estimates that are close to zero, rather

[38]We obtain a similar result when we apply the PBT approach to two-year placebo data periods, formed from adjacent pre-period tax years (Appendix Figure A.11). An additional consideration that might support a linear time trend is that zooming out from the 4-month bandwidth employed in our main California specification provides evidence that maternal labor force participation gradually declines with respect to child birth date (Appendix Figure A.12); a linear specification that accounts for that trend may yield lower bias (even while generating higher variance).

than one that performs well at obtaining estimates that are close to the true effect size. If so, this could bias our estimates towards zero even when the true effect size was non-zero.[39] To assess this concern, we conduct Monte Carlo simulations assuming a true treatment effect that is non-zero. The results suggest that the estimates obtained through our proposed method are unbiased for the set of parameters we consider (Appendix Figure A.13).

Finally, because our main analysis conditions on having worked in the prior year, the time-path of estimates we observe could be shaped by time-varying selection into our sample. In particular, our treatment effect is identified by differences in mothers' work decisions in their last year of credit eligibility, among mothers who chose to work in prior years of eligibility. Yet the credit itself could have shaped those prior-year work decisions. To explore this possibility, we replicate our analysis for the sample of mothers who worked in the year prior to the credit's adoption.[40] For this analysis, the credit's incentive effects do not (by construction) affect inclusion into the sample. Appendix Table A.14 shows that the results from this alternative sample are similar to those obtained from our main analysis.

5.4 Heterogeneity of Estimated Treatment Effects

Income Although the tax credits we study shape labor supply incentives throughout the income distribution, the effects may be particularly strong for lower-income individuals, for whom the credit amounts constitute a larger share of after-tax income. To investigate this possibility, we replicate our main analysis separately for taxpayers who had prior-year income below or above $30,000.[41] We find no evidence of substantial labor force participation effects

[39] In practice, this possibility seems remote given the parameters over which we are optimizing – e.g., there is no reason to expect that a narrower or wider bandwidth would tend to yield estimates that are mechanically closer to zero. The concern would be greater if the treatment effect was present only among observations close to the treatment threshold. This seems unlikely in our context; the labor supply incentives facing a parent whose child turns six one month after the cutoff are the same as those facing a parent whose child turns six two months after the cutoff.

[40] We limit this analysis to states with at least two years of outcome data observed post credit adoption, since otherwise the new selection criterion amounts to mothers who worked in the prior-year, i.e., the selection criterion for our main analysis.

[41] We consider other income splits, corresponding to phase-out ranges for each state credit, when we estimate elasticities in Section 6, below.

for low-income mothers (see Appendix Table A.15).

Marital Status Another potential source of heterogeneity relates to marital status. Both California and Colorado's work requirements were applied at the tax return level; as such, a non-working mother could qualify for the credit if her spouse earned positive income during the tax year. For this reason, work requirements may have a smaller effect on labor force participation for some married as compared to unmarried mothers. Appendix Tables A.16 replicate our main results separately for single and married mothers. We find no systematic treatment effect differences between these groups. The story is similar when we repeat this analysis separately for taxpayers with prior-year income above versus below $30,000 (Appendix Table A.17).[42]

Self-Employment Status A consistent finding from prior research is that self-employed taxpayers respond more strongly to tax incentives, potentially because they have more flexibility to adjust their labor supply (see le Maire and Schjerning (2013); Bastani and Selin (2014); Wilson (2026)).[43] Appendix Table A.18 investigate heterogeneity along this dimension. We find no evidence for an effect for either group, although our results for self-employed taxpayers are somewhat less precise due to the reduction in sample size.

Other Dimensions of Heterogeneity We also explore several other dimensions of heterogeneity that could plausibly mediate maternal responsiveness to the tax credit: urbanicity, maternal college attendance, and the presence of a younger child in the household (Appendix Tables A.19 through A.21). We also consider the specific subgroup (low-income single mothers with lower education) that was found by Schanzenbach and Strain (2023) to respond to

[42]Prior work studying the EITC finds the largest effects among lower income single mothers (e.g. Meyer and Rosenbaum, 2001; Nichols and Rothstein, 2016). Although we find evidence for a marginally significant effect among this subgroup in one state (Vermont), the presence and even direction of estimated heterogeneity is not consistent across states.

[43]An alternative explanation is that the greater observed response exclusively reflects differences in tax reporting behavior; however, the growing use of third-party information reporting for this group may limit this channel in our analysis (see Garin, Jackson and Koustas (2025)).

a change in child tax credit work requirements (Appendix Table A.22). We do not observe systematic evidence of a treatment effect with respect to any of these subgroups.

5.5 Alternative Samples with Non-Working Mothers

Our main sample consists of mothers who, by construction, were already attached to the labor force. As such, our estimates thus far primarily reflect the effect of tax credit eligibility on labor force exit. However, eligibility for these credits could also shape the degree to which non-working mothers enter the labor force. To study both of these flows, we supplement our main analysis with two alternative samples.

The first alternative sample consists of mothers of children enrolled in Medicaid or CHIP at some point during the prior year, regardless of whether or not the mother was working. Summary statistics for this sample are provided in Appendix Table A.23. In each of our six states, roughly two-thirds of mothers in this sample worked during the prior year.

For our California analysis, we additionally replicate our results with a second alternative sample, using administrative data accessed through the U.S. Census Bureau. We use this data to identify the universe of children born in California who turned six around the end of one of our outcome years. Construction of this sample relies on the Social Security Administration's Numident file, the Census Household Composition Key, filed tax returns, and third-party wage and salary income information returns (Form W-2). In the Census sample, we cannot link parents to information returns other than the W-2s, meaning our employment measure is limited to wage earnings and does not consider self-employment. Like the Medicaid sample, the Census sample has the advantage of allowing us to identify children likely residing in California during the relevant outcome years without needing to condition on mothers' prior employment and tax filing.

Table 3 replicates our difference-in-discontinuities analysis for California using the Medicaid and Census samples, and Table 4 replicates our analyses for the other states using the Medicaid sample. In both cases, the analysis is conducted separately for prior-year workers

(Panel A) and prior-year non-workers (Panel B).[44] Consistent with our main sample, we estimate small and statistically insignificant effects for previously working mothers.

In contrast, we find some suggestive evidence that credit eligibility increases labor force participation among mothers who were not working in the prior year. In particular, in Colorado, the estimated effects for previously non-working mothers are larger than for previously working mothers, but neither effect (nor the difference between them) is statistically significant. However, when we pool across all Colorado mothers in the Medicaid sample (whether previously working or not), we estimate that the credit led to a marginally significant 0.77 percentage point increase in labor force participation ($p < 0.10$) (see Appendix Table A.25).

We also find some evidence that credit eligibility increases labor force participation among previously non-working mothers in California. Interestingly, the evidence for this effect is most consistent in the time period after California's work requirement was eliminated. For the Medicaid sample, we estimate positive and statistically significant effects both before and after the elimination of the work requirement. For the Census sample, we do so only after the elimination of the work requirement.[45] Why might credit eligibility increase labor force participation absent a work requirement? A positive effect on labor participation could be due to a positive income effect—e.g., from eliminating financial barriers to work. Alternatively, it could reflect a shift from informal to formal sources of employment associated with one's decision to start filing a tax return in order to claim the credit. Because the positive estimated effects for non-working mothers are not limited to time periods for which the work requirement was in effect, the difference-in-discontinuities results for non-working mothers in California are either zero (Medicaid sample) or positive (Census sample). We interpret these results to suggest that while the availability of the tax credit may increase labor force participation for non-working mothers, the mechanism through which it does so is unlikely to be a work requirement.

[44] See Appendix Tables A.24 and A.25 for pooled results.

[45] For other states without a work requirement, we find a positive and marginally significant effect in Vermont, and no statistically significant effect in Maryland, New Jersey, or Oregon.

Overall, the results from these alternative samples are largely consistent with those from our main sample in that both imply that the credits did not cause previously working mothers to exit the labor force. At the same time, the analyses in this subsection raise the possibility that the credits may have affected the share of non-working mothers entering the labor force.

5.6 Effects on Reported Income and Tax Filing

Our primary measure of labor force participation is based on third party information returns, rather than income reported on the taxpayer's return. This measure has the advantage of being available regardless of whether an individual files a tax return; hence, any effect we observe is likely to represent a real change in labor supply rather than change in what a taxpayer reports on her return.[46] At the same time, measuring income solely based on third-party information could lead us to miss changes in labor income that are not reported by third parties – either because no third party is required to report it or because a third party is required to report it but fails to do so.[47] A taxpayer who earns income that does not appear on an information return may be particularly inclined to report it when doing so qualifies her for a state credit.

We replicate our main analyses using reported income as the outcome in Appendix Tables A.26 and A.27. Specifically, Panels A through C of these tables respectively consider the effects of age-eligibility on whether the mother (along with her spouse, if married and filing a joint return) reported any income from wages, any income from self-employment, or any income from either wages or self-employment. Each outcome takes a value of zero if the mother did not file a tax return. Across measures, we find no evidence that the removal of California's work requirement led to a reduction in reported income. We observe similar results for the other state child tax credits for which reported income data is available after the credit is adopted (Colorado, New Jersey, and Vermont). We interpret these results

[46] Garin, Jackson and Koustas (2025) find that some taxpayers increase reported self-employment income to maximize tax benefits like the CTC and EITC without actually increasing their labor incomes.

[47] An example of labor income unlikely to show up on third party information returns are the payments a sole proprietor receives from payors below the minimum reporting threshold of $600 per year.

to suggest that the estimates based on information returns are unlikely to be obscuring reductions in labor income.

We next study the effect of the state tax credits on tax filing. We find that eliminating the California's work requirement leads to a 0.4 percentage point increase in the filing rate (Appendix Table A.28). Perhaps surprisingly, this filing effect appears to be only partly driven by an increase in filing among taxpayers without earned income (i.e., those newly eligible for the expanded credit), as suggested by the increase in the share of taxpayers filing and reporting positive earnings (Panel C). It may be that eliminating the work requirement increases tax filing by simplifying the credit eligibility rules and thereby increasing the perceived benefit to filing a return—a "woodwork effect" (c.f., Anders and Rafkin, 2026). However, we do not observe evidence of an effect on tax filing when we evaluate the other state tax credits that were in place for tax year 2022 (Appendix Table A.29).

Finally, note that California's work requirement provides an incentive for individuals who would otherwise drop out of the labor force to report at least a small amount of earned income, which would allow them to qualify for the full benefit. We investigate the degree to which taxpayers respond to this incentive by exploring the effect of the credit on the number of taxpayers reporting very low but positive incomes (Appendix Table A.30).[48] We observe a small but statistically significant effect of California's credit during the work requirement period on the share of taxpayers who report very low earned income (amounts below $10,000). Interestingly, Appendix Figure A.14 provides suggestive evidence that this (small) effect stems from already-working taxpayers who reduce the income they report in response to the credit's availability or its phase-out (i.e., an intensive-margin income or substitution effect). In particular, age-eligibility for the credit appears to increase the share of taxpayers reporting very low incomes and reduce the share of taxpayers reporting other positive incomes (Panels A and B). Consistent with this effect being driven by the credit's availability or phase-out,

[48] A potential concern with this analysis is that the taxpayer may report the income on a state but not federal return. In principle, taxpayers are not required to file a federal return to claim the credit. In practice, we expect the vast majority of state filers to report their income on a federal return due to the design of tax preparation software and the modern electronic filing system.

we find no evidence that the share of taxpayers reporting very low incomes is affected by the elimination of the work requirement (Panel C).[49]

6 Labor Supply Elasticities

To facilitate extrapolating our results to other policy settings, we use our estimates to calculate extensive-margin substitution and income elasticities. Doing so requires accounting for the mix of income and substitution effects generated by each type of child tax credit.

First, consider a credit with a work requirement. For individuals whose income from work would fall below the start of the income phase-out, the credit raises the return to work. At the same time, the credit does not affect the after-tax income associated with not working. The credit therefore generates a pure substitution effect with respect to labor force participation. For individuals with income within the phase-out region, the nature of the labor force participation incentive effect is the same, but the magnitude is lower. In contrast, the credit generates no extensive-margin labor supply incentive effects for individuals whose income from work would fall above the top of the credit's phase-out range.[50]

The extensive-margin incentive effects are quite different for a credit that is not conditioned on work. For individuals whose income from work would fall below the start of the credit's income phase-out, the credit generates an extensive-margin income effect, in that it raises individuals baseline after-tax income whether or not they work. In contrast, for individuals whose income from work would be above the start of the credit's income phase-out, the credit generates a *negative* extensive-margin substitution effect—the credit increases the after-tax income associated with not working compared to working. This reduction in the return to work is larger for individuals with income above the top of the phase-out range—who

[49] In a related setting, Cook and East (2026) find that SNAP recipients do not slightly increase work effort to maintain benefit eligibility when faced with work requirements.

[50] Although not our focus, the presence of an income phase-out generates *intensive-margin* substitution effects—taxpayers with income above the start of the phase-out face an incentive to reduce their income—and income effects—working taxpayers with higher after-tax income may reduce their labor income to consume more leisure.

qualify for no credit—compared to those with income within the credit phase-out range.

Considering these varying incentive effects, to calculate the extensive-margin substitution elasticity, we draw on the estimated labor force participation effects for two subgroups of taxpayers for whom the credit generates a pure substitution effect with respect to the extensive-margin of labor supply: (1) individuals facing a work requirement credit whose prior-year income was positive but below the bottom of the credit's income phase-out range; and (2) individuals facing a credit without a work requirement whose prior-year income was above the top of the credit's income phase-out range. We scale the estimated labor force participation effect for each of these groups by the percent change in the return to work induced by the applicable credit, accounting for pre-tax income, federal and state income and payroll taxes (calculated using NBER's TAXSIM program), and SNAP benefits. Details are provided in Appendix C.

The estimates used to construct the extensive-margin substitution elasticity are reported in Appendix Table A.31. On average, we estimate that the credits with work requirements increased the return to work by an average of 6.8% for taxpayers with incomes below their phase-out ranges, whereas the unconditional credits reduced the return to work by an average of 1% for taxpayers with incomes above their phase-out ranges. Scaling our estimated labor force participation effects by the corresponding percentage changes in the return to work yields an average extensive-margin substitution elasticity of 0.047.[51]

This extensive margin elasticity is smaller in magnitude than the elasticities derived from many prior studies of the EITC and CTC. A recent National Academies review of this literature reports substitution elasticities ranging from 0.42 to 2.05 for the EITC and from 0.27 to 0.50 for the CTC (National Academies of Sciences, 2026).[52] Our estimate is

[51]In particular, the substitution elasticity for each state is obtained by scaling the estimated labor force participation effect (in percentage terms) by the average change in the return to work for the corresponding state (also in percentage terms). Taking a population-weighted average of the state-specific elasticities yields an average substitution elasticity of 0.050 for the states with conditional credits and 0.019 for the states with unconditional credits. Averaging again across both credit types yields the 0.047 elasticity reported in the text.

[52]The largest estimated responses to the EITC are for low-income single mothers. In our analysis, we do not observe systematically larger participation effects for this group.

more consistent with research studying parental responses to the federal CTC expansion in 2021, which finds substitution elasticities ranging from 0 to 0.34 (National Academies of Sciences, 2026). Because the work-contingent credits raise the return to work by an average of 6.8%, the elasticity range would imply employment increases of 0% to 14% under the work-contingent regime—or corresponding reductions from eliminating that incentive. The estimated labor force participation effects we observe in both work requirement states (California and Colorado) are at the very low end of this range.[53]

We similarly calculate extensive-margin income elasticities by scaling our estimated labor force participation effects by the percentage change in after-tax income from not working (Appendix Table A.32). Absent the credit, we assume that non-workers receive benefits only from SNAP, abstracting from smaller safety net programs like TANF or non-market income such as charitable support or gifts from family members. As described above, for this exercise we rely on the estimated effects for the credits that are not conditioned on work, from taxpayers with incomes below the start of the applicable phase-out range. Across states, we estimate that the credits increase the after-tax income associated with non-working by a population-weighted average of 130%. Scaling our point estimates by this change yields a very small extensive-margin income elasticity, less than 0.01.

7 Conclusion

We study the labor force participation consequences of conditioning child tax benefits on work. The question of whether to limit eligibility for child tax benefits to taxpayers with labor income is a central issue in current policy debates at both the federal and state levels. Our empirical design identifies this effect for mothers of young children. The results of our analysis suggest that requiring these mothers to work in order to receive benefits is unlikely to

[53]In particular, the labor force participation effects reported in Appendix Table **??** correspond to 95% confidence intervals with upper endpoints of 0.65 percentage points (California) and 1.11 percentage points (Colorado). Scaling by the control means yields implied labor participation effects of 0.7% for California and 1.2% for Colorado.

be an effective means of increasing labor force participation in the policy setting we consider.

Our most direct evidence for the effect of eliminating a work requirement comes from California. A potential concern with drawing broader conclusions from California's experience is that the design of its credit differs from the design of other state and federal child tax credits, particularly in terms of the starkness of its work requirement. However, the results from the other states we study suggest that the observed effects of California's credit are not due to its distinctive design. In particular, we observe comparable results from the California credit prior to its work requirement being eliminated as from the Colorado credit—which has a more conventionally designed work requirement and benefit schedule. Similarly, we observe comparable results from the California credit after the elimination of its work requirement as from the other states that never conditioned benefit eligibility on work. The similarity of the effects in California to those in other states increases our confidence in extrapolating from California's experience to other policy settings.

That being said, there are several related reasons to be cautious before applying our results to policy reforms that would dramatically alter the relationship between benefit program eligibility and work in the United States. The effect of child tax credit work requirements depends on how many parents are on the margin of employment after accounting for the full set of policies that shape their financial return to work. The state credits we study operate in the shadow of the EITC and the large work incentives it generates. Reforms that eliminate or substantially modify the design of that credit could therefore produce very different results. Relatedly, our estimated (non)response to unconditional child tax credits may understate how much taxpayers would respond to much larger unconditional income grants that come closer to covering the cost of living. For example, Balakrishnan et al. (2024) and Vivalt et al. (2024) document negative employment effects from a $500 to $1,000 *per month* universal basic income pilot, although the effects are smaller for parents than non-parents. In contrast, Jones and Marinescu (2022) and Akee et al. (2010) find no labor supply effects of Alaska's guaranteed income program or government transfers tied to casino profits, respec-

tively, which provide unconditional benefits that are closer in size to the credits we study. In light of these considerations, our main conclusion is that marginally reducing pro-work incentives in child tax benefits is unlikely to substantially reduce labor force participation, especially if meaningful work requirements continue to exist for other safety net programs like the EITC.

Finally, with respect to methodology, our proposed placebo-based tuning approach is potentially applicable to a range of settings in which researchers use RD designs to evaluate the effects of a policy. The main constraint is the presence of a placebo data set in which the policy is not in effect and that otherwise approximates the relevant bias-variance trade-offs for the researcher's actual sample of interest. Subject to this requirement, there are a number of settings in which such a placebo could potentially be available. First, the placebo could consist of a time period in which the policy is not in effect (our setting) or in which the discontinuity threshold occurs at a different cutoff point – e.g., a change in the birth date cutoff to enter school or participate in a social benefit program. Second, the placebo could consist of other jurisdictions for which the policy is not in effect. Third, the placebo could consist of individuals within the applicable time period and jurisdiction but who are not eligible for the policy being studied. When a suitable placebo is available, our results illustrate how placebo-based tuning may yield substantial performance gains for RD designs.

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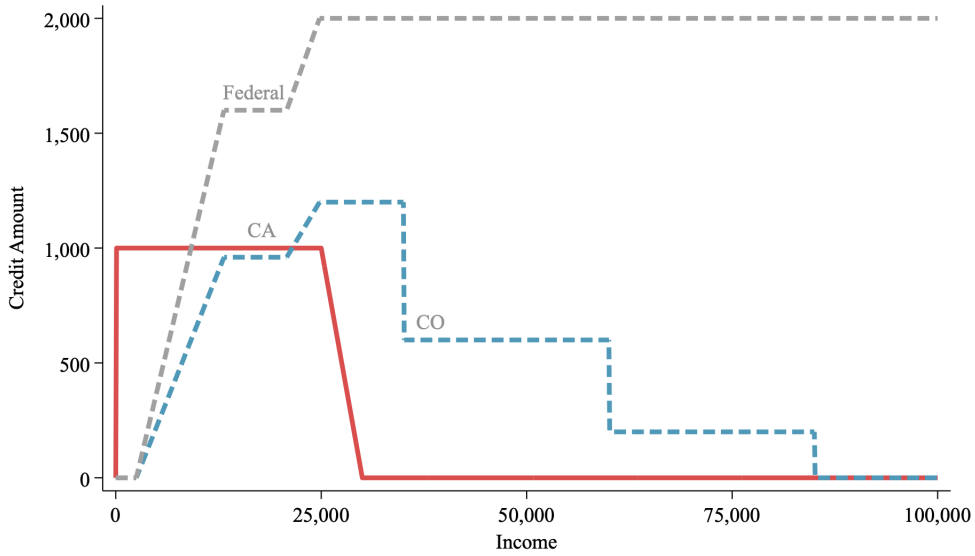
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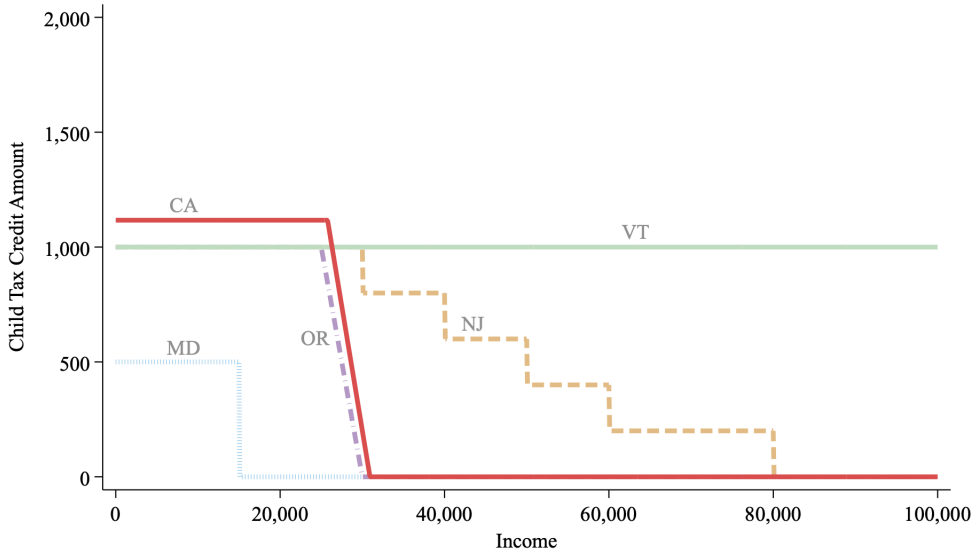
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Figure 1: Benefit Schedules: Federal and State CTCs

(a) Work Requirement

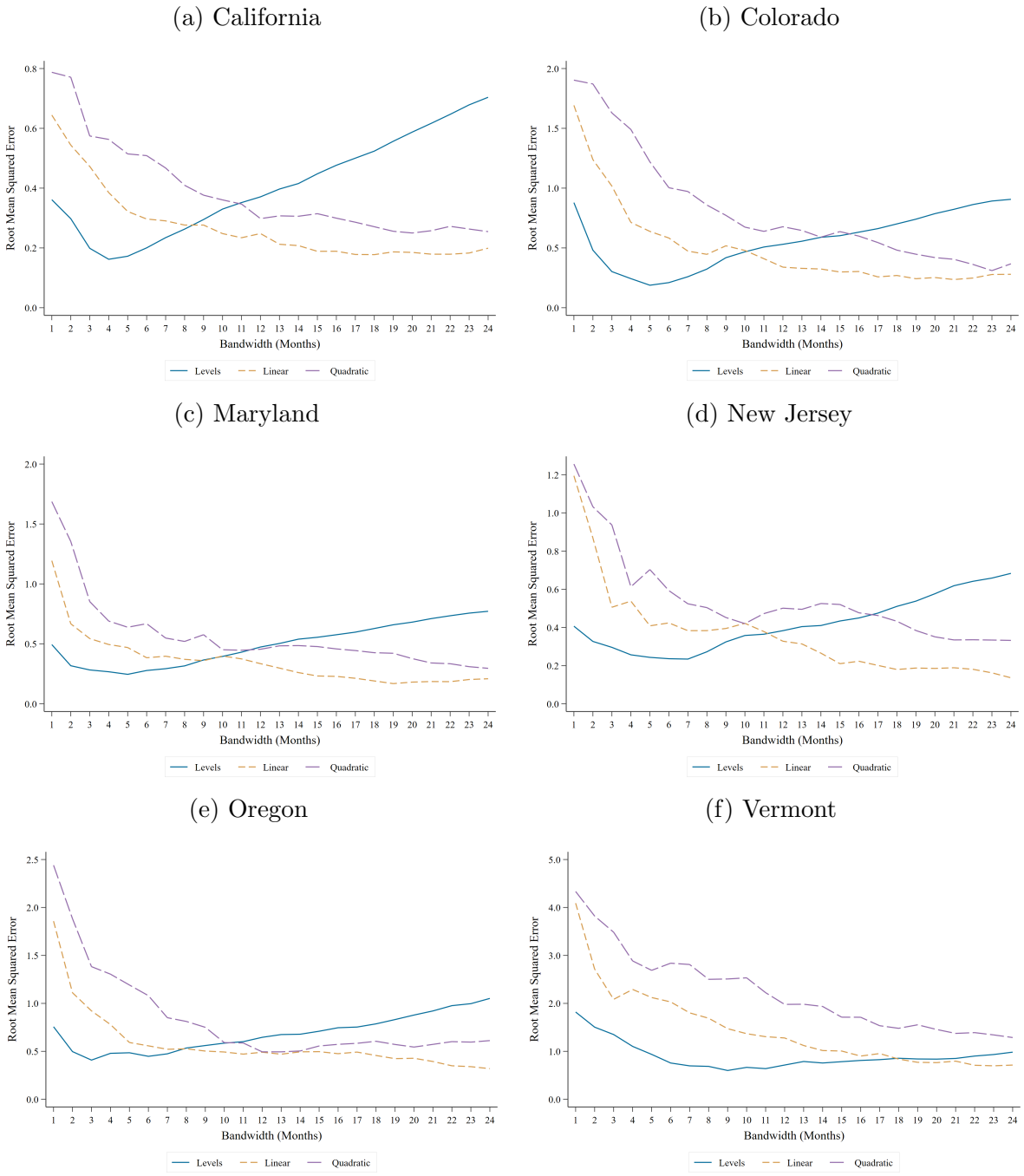


(b) No Work Requirement



Notes: The figure presents the benefit schedule for the federal CTC and the six states that provide child tax credits for parents of children under six. Panel A shows the benefit schedule for credits with work requirements: the federal CTC, the Colorado credit, and the California credit prior to 2022. Panel B shows the benefit schedule for credits without work requirements: Maryland, New Jersey, Oregon, Vermont, and California after the work requirement was removed. Vermont's credit phase-out is not pictured; it occurs between incomes of $125,000 and $175,000. Dollar amounts are for tax year 2023, except for the California credit with work requirement, for which the reported dollar amount is for 2020. Each calculation assumes a single taxpayer who claims a single qualifying child who is under age six and reports only earned income.

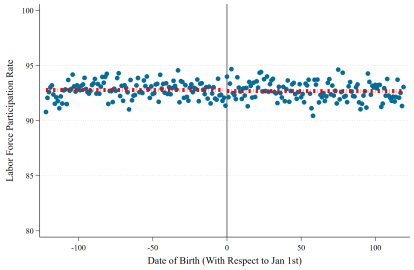
Figure 2: Placebo-Based Tuning Analysis



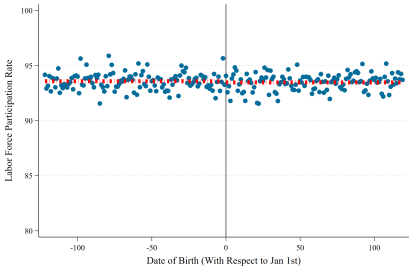
Notes: The figure reports the results of the Placebo-Based Tuning Analysis for California, Colorado, Maryland, New Jersey, Oregon, and Vermont described in Section 4.2. It compares the root mean squared error (RMSE) of the distribution of estimated (placebo) effects for different empirical specifications. Each placebo effect is obtained from estimating the pseudo-effect of age-eligibility for the given state's child tax credit following equation (1), for each year (other than 2021) from 2005 until the introduction of the credit, in a specification using a local polynomial with the specified bandwidth and order, with uniform weights, and with no "donut" exclusion. Each estimate is obtained from a sample composed of mothers whose focal child turns six within the specified bandwidth (i.e., the number of months on either side of the end of the year). The reported RMSE corresponds to the square root of the average (across years) of the square of the estimated coefficients from each year. The purple, long dashed line includes a quadratic polynomial in date of birth; the yellow, short dashed line includes a linear trend in date of birth; and the solid blue line consists of a simple comparison of means of the age-eligible and age-ineligible groups of mothers.

Figure 3: Labor Force Participation by Child's Date of Birth

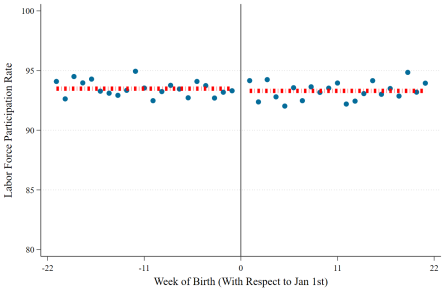
(a) California, Work Req.



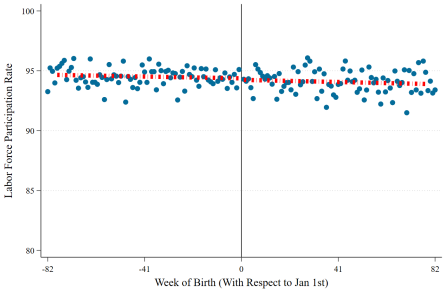
(b) California, No Work Req.



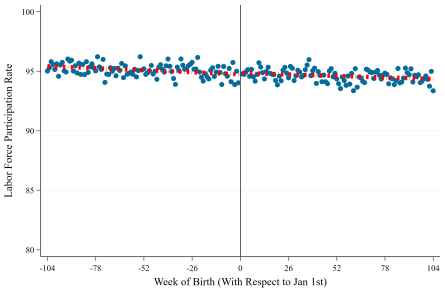
(c) Colorado



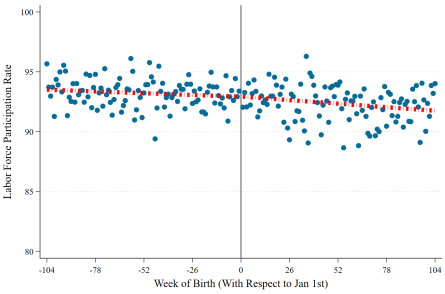
(d) Maryland



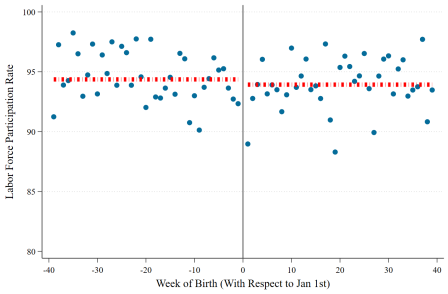
(e) New Jersey



(f) Oregon



(g) Vermont



Notes: The figure reports mean labor force participation rates binned by child's week of birth for recently working mothers. The sample consists of individuals who had positive wage or self-employment income reported on third-party information returns during the previous tax year and who has a child whose sixth birthday falls within the PBT-selected bandwidth for the state. The dashed lines correspond to the state's local polynomial, estimated separately for dates before versus after the turn of-the-year (normalized to zero).

Table 1: California Credit Eligibility and Labor Force Participation

	(1)	(2)	(3)
	Work Req. (2019-2020)	No Work Req. (2022-2023)	Diff-in-Disc [(2) - (1)]
Age-eligible	-0.113	-0.114	-0.001
	(0.103)	(0.099)	(0.143)
95% CI	[-0.32, 0.09]	[-0.31, 0.08]	[-0.28, 0.28]
Control Mean	92.8	93.6	93.2
Observations	251,699	244,828	496,527

Notes: The table reports the estimated effect of eligibility for the California YCTC separately during the work requirement period (Column 1) and period without a work requirement (Column 2). Column 3 corresponds to the difference in estimated effects between Columns 1 and 2. The outcome in each column is an indicator for having positive wage or self-employment income reported on third-party information returns; units are percentage points (0-100). The sample consists of recently working mothers in California whose youngest child's sixth birthday falls within the four-month window around the end of the specified year. The control mean corresponds to mothers of age-ineligible children. Parentheses report heteroskedasticity-robust standard errors.

Table 2: Other State Credit Eligibility and Labor Force Participation

	(1)	(2)	(3)	(4)	(5)
	Colorado	Maryland	New Jersey	Oregon	Vermont
Age-eligible	0.032	-0.052	-0.033	-0.069	-0.535
	(0.184)	(0.237)	(0.128)	(0.330)	(0.393)
95% CI	[-0.33, 0.39]	[-0.52, 0.41]	[-0.28, 0.22]	[-0.72, 0.58]	[-1.30, 0.23]
Control Mean	93.2	94.8	95.3	93.4	94.7
Observations	74,873	138,541	441,346	91,310	13,746
Bandwidth (Months)	5	19	24	24	9
Polynomial Order	0	1	1	1	0
Work Req	Yes	No	No	No	No
Max Credit ($)	1,200	500	500-1,000	1,000	1,000
Years	2022-2023	2023	2022-2023	2023	2022-2023

Notes: The table reports the estimated effect of eligibility for the state child tax credits for states other than California. The outcome in each column is an indicator for having positive wage or self-employment income reported on third-party information returns; units are percentage points (0-100). The sample consists of recently working mothers in the specified state who have a child whose sixth birthday falls within the specified bandwidth around the end of the specified year. Polynomial order 0 refers to a simple comparison of means; polynomial order 1 refers to a specification that includes a linear trend in the child's date of birth. The bandwidth and polynomial for each state is determined by applying the placebo-based tuning approach to that state, as described in the text. Work req indicates whether eligibility for the credit is limited to parents with income from work. The control mean corresponds to mothers of age-ineligible children. Parentheses report heteroskedasticity-robust standard errors.

Table 3: California Credit Eligibility and Labor Force Participation by Prior-Year Employment

	Medicaid Sample			Census Sample		
	(1)	(2)	(3)	(4)	(5)	(6)
	Work Req. (2019-2020)	No Work Req. (2022-2023)	Diff-in-Disc [(2) - (1)]	Work Req. (2019-2020)	No Work Req. (2022-2023)	Diff-in-Disc [(5) - (4)]
Panel A: Worked in Prior Year						
Age-eligible	-0.200	-0.241	-0.040	0.029	0.021	-0.008
	(0.167)	(0.166)	(0.235)	(0.097)	(0.088)	(0.130)
95% CI	[-0.53, 0.13]	[-0.57, 0.08]	[-0.50, 0.42]	[-0.16, 0.22]	[-0.15, 0.19]	[-0.26, 0.25]
Control Mean	90.3	90.8	90.5	92.9	92.7	92.8
Observations	127,145	123,325	250,470	280,000	350,000	630,000
Panel B: Did Not Work in Prior Year						
Age-eligible	0.645*	0.823*	0.178	0.025	0.655***	0.630*
	(0.310)	(0.343)	(0.462)	(0.217)	(0.190)	(0.288)
95% CI	[0.04, 1.25]	[0.15, 1.49]	[-0.73, 1.08]	[-0.40, 0.45]	[0.28, 1.03]	[0.07, 1.19]
Control Mean	19.1	22.0	20.5	16.3	18.9	17.8
Observations	65,089	59,088	124,177	120,000	170,000	290,000

Notes: The table replicates the analysis from Table 1 separately for mothers who worked during the prior year (Panel A) and for mothers who did not work during the prior year (Panel B). A mother is treated as having worked during the prior year if she has positive wage or self-employment income for that year reported on third-party information returns. Columns 1-3 report results from the Medicaid sample, which is comprised of mothers of children enrolled in Medicaid in the state of California at any point during the prior year. Columns 4-6 report results from the Census sample, which is comprised of birth mothers of children born in California; birth mothers are identified from the Census Household Composition Key (CHCK) and birth location is according to the Social Security Administration (SSA) Numident file. For the Medicaid sample, the outcome in each column is an indicator for having positive wage or self-employment income reported on a third-party information return. For the Census sample, the outcome is an indicator for positive wage income reported on a W-2 form. All counts and estimates are rounded per Census's disclosure rules governing administrative records. The Census Bureau has reviewed results from the Census sample to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data used to produce this product (Data Management System (DMS) Number: P-7503840, Disclosure Review Board (DRB) approval number: CBDRB-FY24-SEHSD003-066.) * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 4: Other State Credit Eligibility and Labor Force Participation by Prior-Year Employment

	(1)	(2)	(3)	(4)	(5)
	Colorado	Maryland	New Jersey	Oregon	Vermont
Panel A: Worked in Prior Year					
Age-eligible	0.081	-0.155	0.096	-0.207	-0.723
	(0.317)	(0.407)	(0.242)	(0.507)	(0.670)
95% CI	[-0.54, 0.70]	[-0.95, 0.64]	[-0.38, 0.57]	[-1.20, 0.79]	[-2.04, 0.59]
Control Mean	90.9	92.2	92.8	91.3	91.7
Observations	32,744	67,485	178,270	49,693	7,052
Panel B: Did Not Work in Prior Year					
Age-eligible	0.959	-1.400	-0.352	-0.074	3.055
	(0.727)	(1.010)	(0.603)	(1.074)	(1.562)
95% CI	[-0.47, 2.38]	[-3.38, 0.58]	[-1.53, 0.83]	[-2.18, 2.03]	[-0.01, 6.12]
Control Mean	26.2	20.8	24.1	22.2	23.7
Observations	14,842	24,466	77,164	23,488	3,091

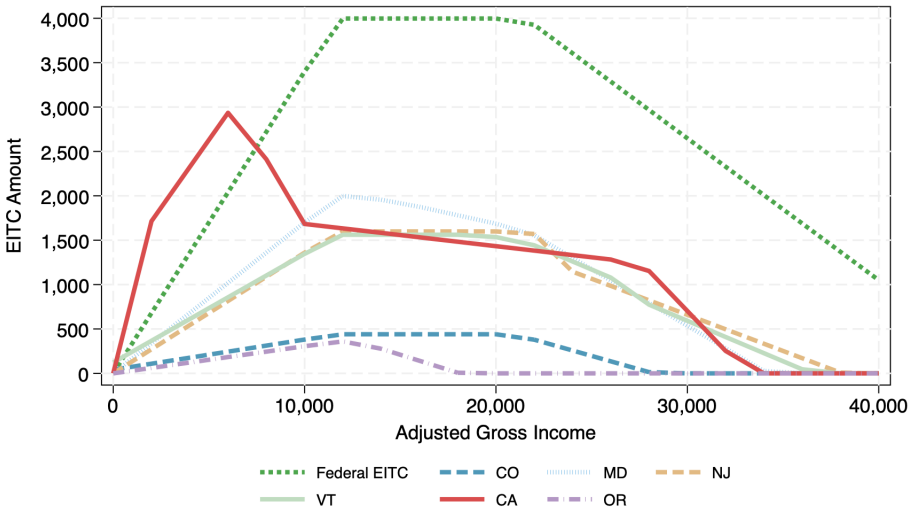
Notes: The table replicates the analysis from Table 2 separately for mothers who worked during the prior year and for mothers who did not work during the prior year. The analysis is conducted using the Medicaid sample, which is comprised of mothers of children enrolled in Medicaid in the specified state at any point during the prior year. A mother is treated as having worked during the prior year if she has positive wage or self-employment income for that year reported on a third-party information return.

Online Appendix to Work Requirements and Child Tax Benefits

Jacob Goldin Tatiana Homonoff Neel Lal
Ithai Lurie Katherine Michelmore Matthew Unrath

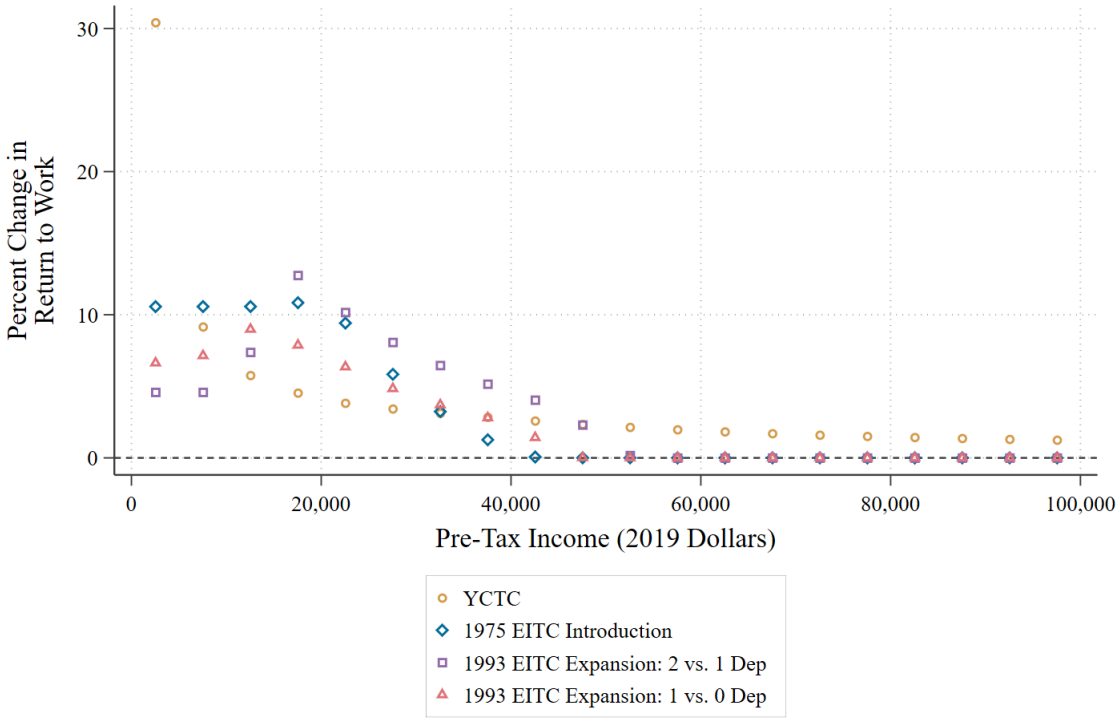
A Appendix Tables and Figures

Figure A.1: Benefit Schedules: Federal and State EITCs



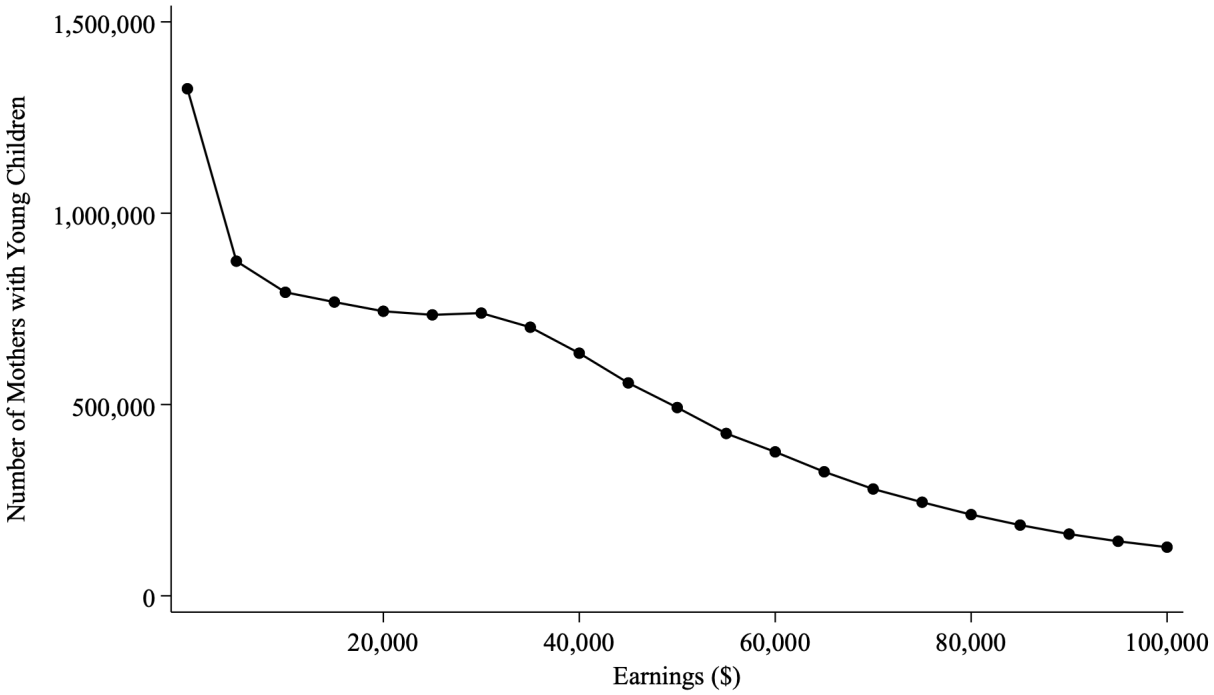
Notes: The figure presents the benefit schedule for the federal and state EITCs for the six states in our sample. Dollar amounts are for tax year 2023. Each calculation assumes a single taxpayer who claims a single qualifying child who is under age six and reports only earned income.

Figure A.2: Percent Change in Return to Work



Notes: The figure plots the percentage change in the return to work for a range of policy reforms. For each reform, the percentage change in the return to work is calculated as the difference between the post-reform return to work minus the pre-reform return to work, divided by the pre-reform return to work. The return to work at a given level of pre-tax income is defined as the after-tax income at the specified pre-tax income minus the after-tax income for which the taxpayer would be eligible if the taxpayer were not to work. After-tax income is calculated net of federal income taxes and payroll taxes, using the NBER TAXSIM model. The plotted points are derived from the calculated percentage change in the return to work at $100 intervals, averaged into $5,000 bins. Pre-tax income is based on 2019 dollars; earlier policy changes are inflation-adjusted to 2019 dollars for comparability. Unless otherwise specified, tax liability is calculated assuming an unmarried taxpayer with one child. The figure considers three policy reforms: the elimination of California's YCTC work requirement, the introduction of the EITC in 1975, and the expansion of EITC generosity from the Omnibus Budget Reconciliation Act of 1993 (OBRA93). For OBRA93, we separately plot the change in the return to work for taxpayers with one child versus no children, and for taxpayers with two children versus one. For each reform, the pre-reform return to work is calculated based on the tax law in place for the tax year prior to the reform; we assume that the only change in policy is the specified reform, so we do not account for other contemporaneous changes in policies such as changes in statutory tax rates in the year that the reform first applies.

Figure A.3: Number of Mothers with Young Children by Earnings Level

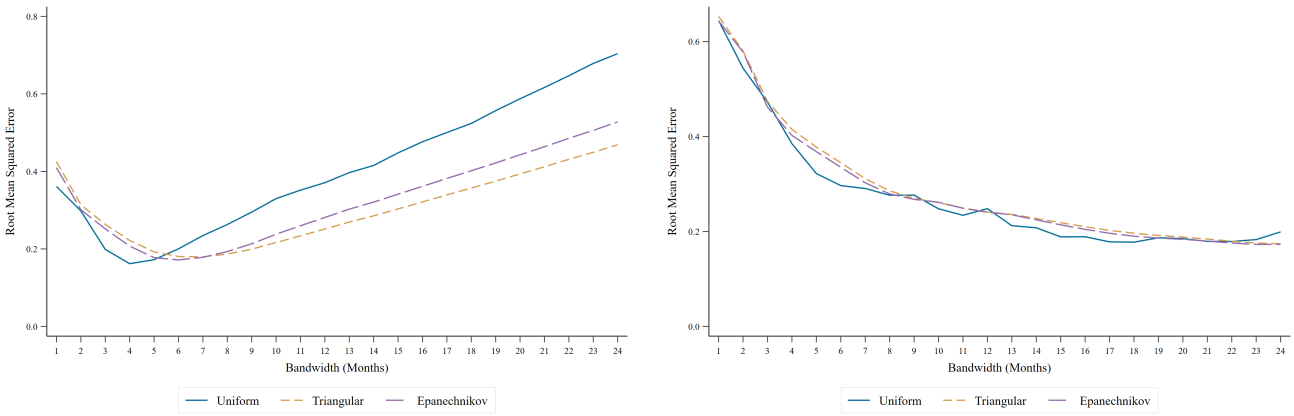


Notes: The figure plots the number of mothers nationwide with children under the age of six by earnings level. Child age is measured as of the close of 2023. Earnings correspond to wage or self-employment income reported on third-party information returns for tax year 2023. The sample is limited to individuals who received one or more information return; hence non-earners are not included in the analysis. Each plotted dot corresponds to an income bin of $5,000.

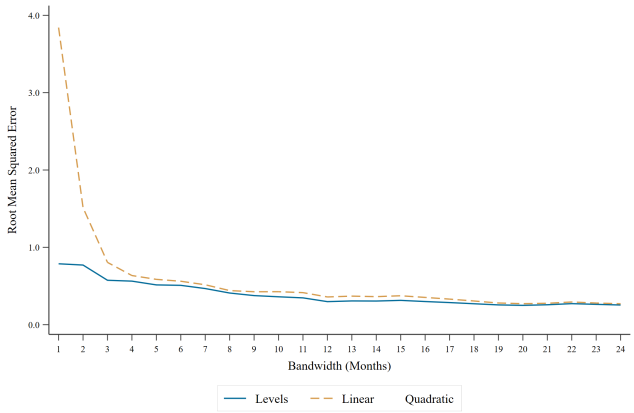
Figure A.4: Placebo-Based Tuning Analysis: Optimal Kernel

(a) Levels

(b) Linear

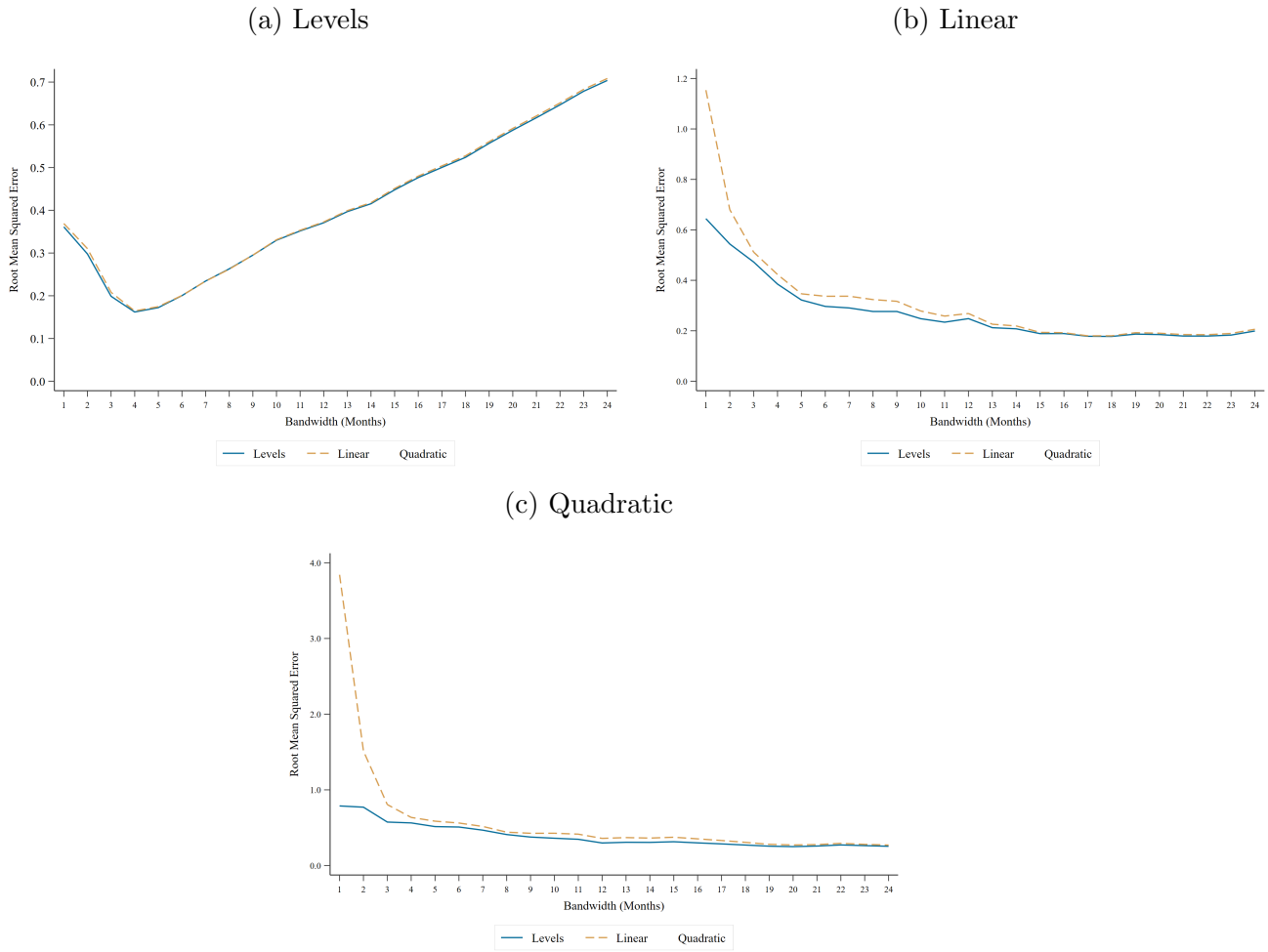


(c) Quadratic



Notes: The figure reports the results of the Placebo-Based Tuning Analysis to determine the optimal local polynomial kernel for the regression discontinuity specification for our California analysis, as described in Section 4.2. The figure repeats the analyses in Figure 2, but varies the choice of kernel. The kernels considered are uniform (blue/solid), triangular (yellow/short dash), and Epanechnikov (purple/long dash). The three panels correspond to a simple comparison of means (Panel A), a linear trend in child's date of birth (Panel B), or a quadratic trend in child's date of birth (Panel C).

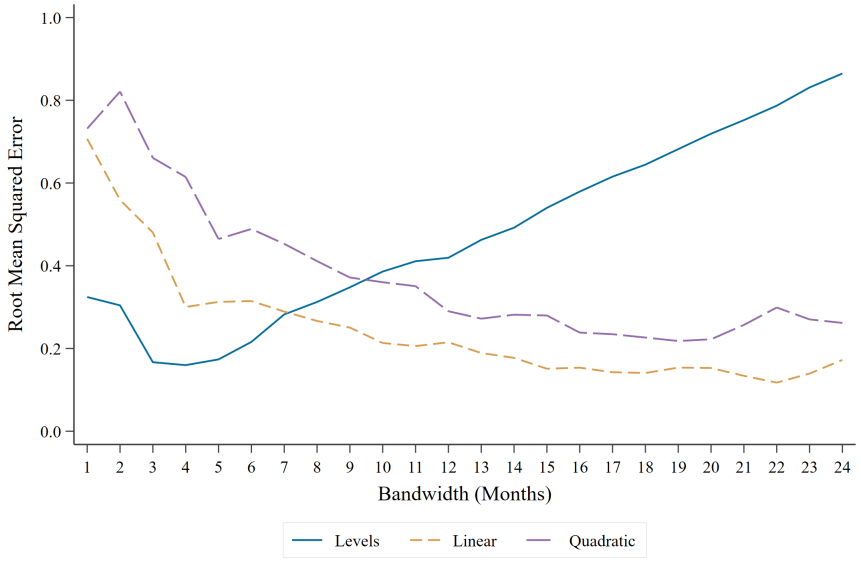
Figure A.5: Placebo-Based Tuning Analysis: Donut Exclusion



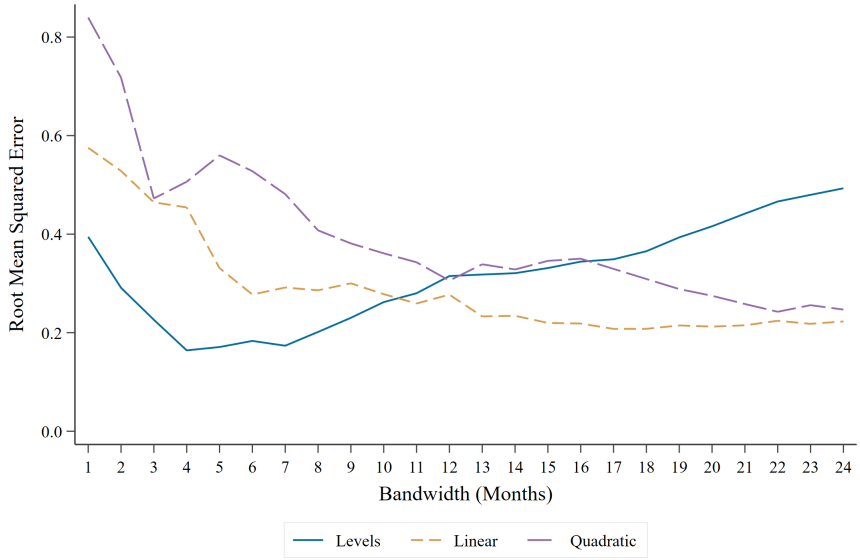
Notes: The figure reports the results of the Placebo-Based Tuning Analysis to determine whether to exclude an 8-day "donut" of births around the turn of the year for the regression discontinuity specification for our California analysis, as described in Section 4.2. The figure repeats the analyses in Figure 2, but varies whether a donut of births is excluded. The dashed yellow lines reflect analyses that exclude the donut of births; the solid blue lines reflect analyses that do not. The three panels correspond to a simple comparison of means (Panel A), a linear trend in child's date of birth (Panel B), or a quadratic trend in child's date of birth (Panel C).

Figure A.6: Placebo-Based Tuning Analysis by Pre-Period Segment

(a) 2005-2011

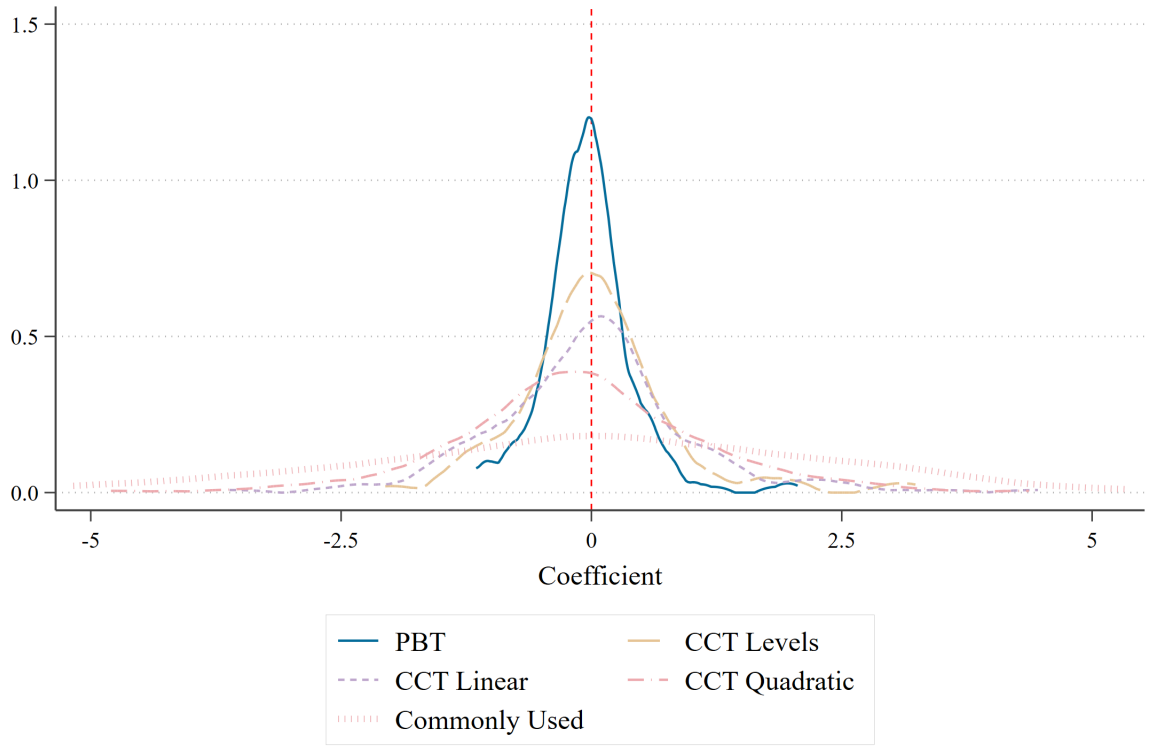


(b) 2012-2018



Notes: The figure replicates the analysis in Figure 2 separately for the first and second halves of the pre-period. The purple, long dashed line includes a quadratic polynomial in date of birth; the yellow, short dashed line includes a linear trend in date of birth; and the solid blue line consists of a simple comparison of means of the age-eligible and age-ineligible groups of mothers. Panel A reports results for 2005-2011; Panel B reports results for 2012-2018.

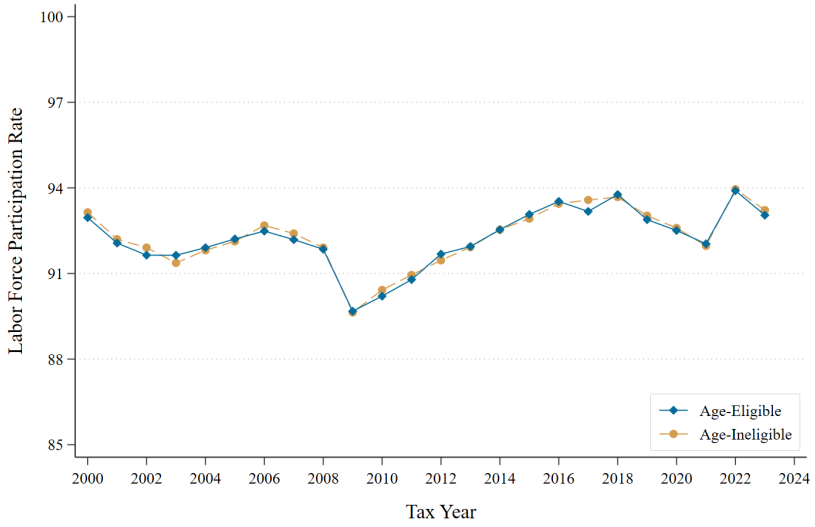
Figure A.7: Distribution of Pseudo-Treatment Effects for Placebo State-Years by Specification Selection Method



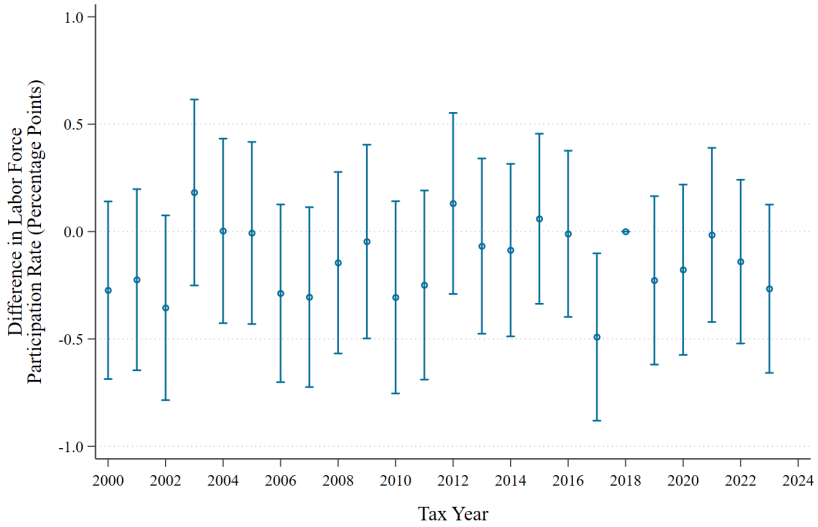
Notes: The figure reports the results of a placebo exercise comparing alternative methods for selecting the RD specification. The placebo states consist of each state that did not provide a young child tax credit during our sample period (i.e., all states and the District of Columbia other than California, Colorado, Maryland, New Jersey, Oregon, and Vermont). For each placebo state, we apply the placebo-based tuning analysis to 2005-2018 to select that state's optimal RD specification. We then use the optimal specification from each state to obtain estimates for that state for each outcome year (2019, 2020, 2022, and 2023). We separately obtain a distribution of estimates for each placebo state-year based on the optimal bandwidth selection method proposed in Calonico, Cattaneo and Titiunik (2014), implemented using the RDRobust Stata package with the package default (triangular) kernel. Because that method takes the local polynomial order as given, we alternatively apply the CCT method to local polynomials of order 0 (levels), 1 (linear), and 2 (quadratic). The "commonly used" distribution refers to placebo state year estimates drawn from an RD specification with a one-month bandwidth, a local linear polynomial, and a donut exclusion of births 8 days around the turn-of-the-year.

Figure A.8: Labor Force Participation by Eligibility for the California Credit

(a) Labor Force Participation Rate

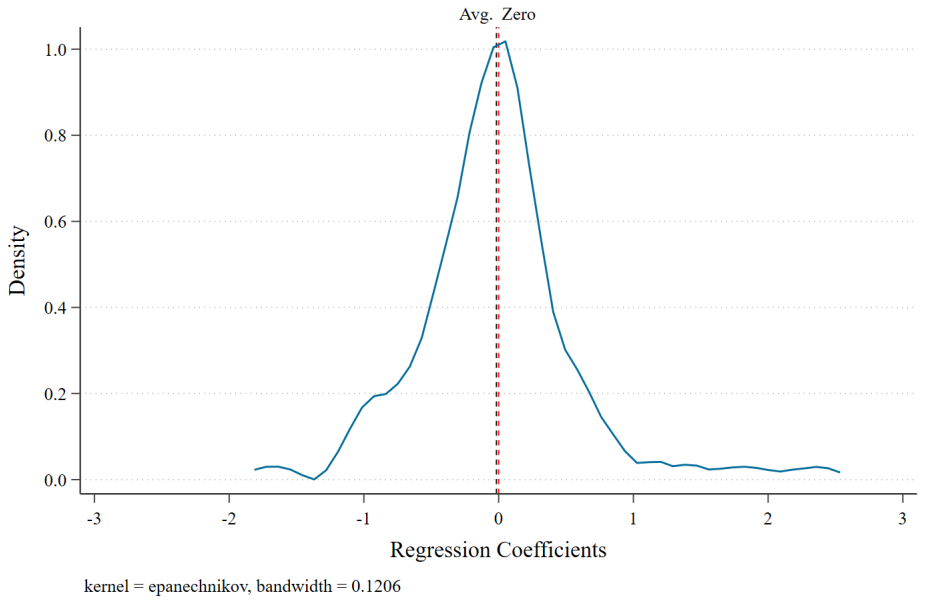


(b) Differences by Age-Eligibility



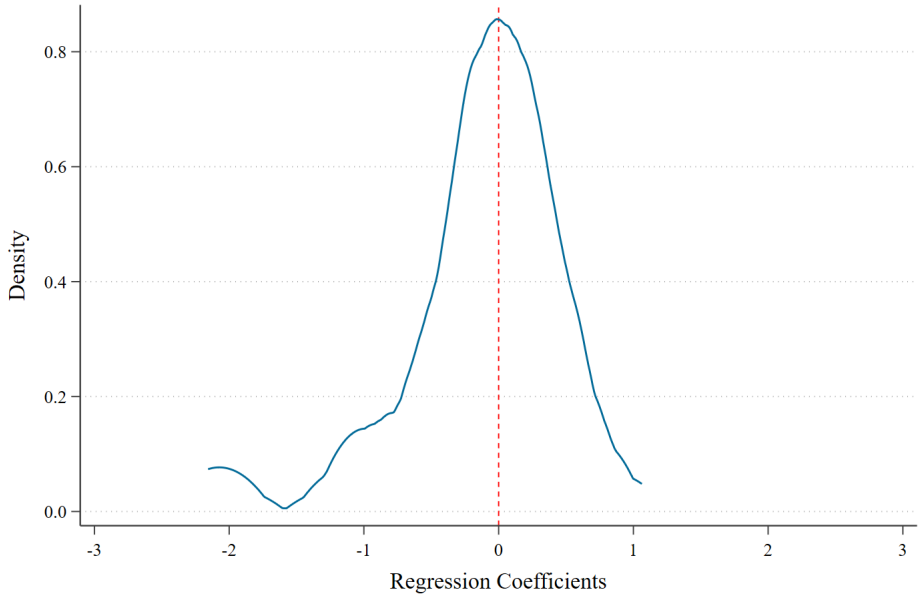
Notes: Panel A reports labor force participation rates by year for recently working California mothers by age-eligibility for the credit. Mothers whose youngest child turns six during the last four months of a year (age-ineligible group) are reported in yellow; mothers whose youngest child turns six during the first four months of the subsequent year (age-eligible group) are reported in blue. The sample consists of individuals who had positive wage or self-employment income reported on third-party information returns during the previous tax year. Panel B reports the differences in maternal labor force participation rates between age-eligible and age-ineligible groups. Coefficient estimates are reported in percentage points (0-100). Bars represent estimated 95% confidence intervals, derived from heteroskedasticity-robust standard errors.

Figure A.9: Distribution of Pseudo-Effects for Placebo State-Years



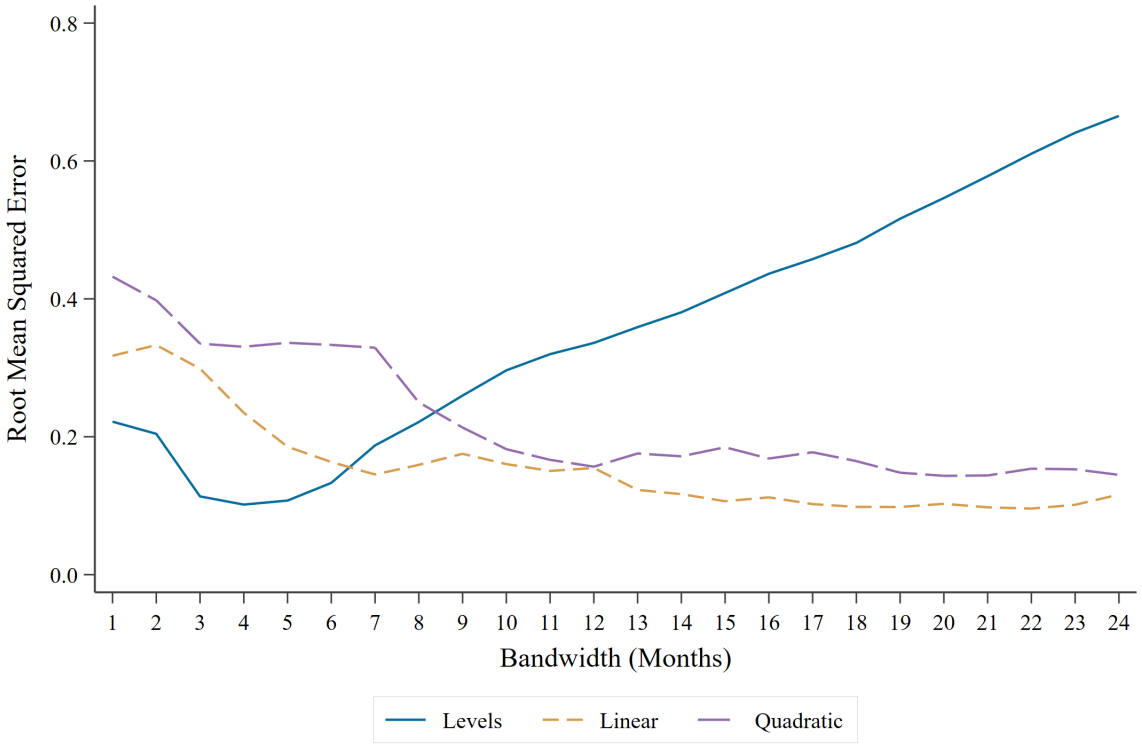
Notes: The figure reports the distribution of estimated pseudo-treatment effects for mothers in state-years that do not have a young child tax credit during the sample period using the PBT-optimal specification for the California analysis. For each state and year, the estimated effect is obtained by comparing labor force participation in the given year among mothers who worked in the given state in the year prior to the given year, and whose youngest child's sixth birthday falls within the four-month window around the turn of the given year. The analysis includes coefficients from 2019, 2020, 2022, and 2023, and from all states and the District of Columbia other than California, Colorado, Maryland, New Jersey, Oregon, and Vermont. The figure plots a kernel-density figure with an epanechnikov kernel and bandwidth of 0.1206. The dashed vertical lines denote the sample mean of the distribution (black) and zero (red), as labeled in the figure.

Figure A.10: Distribution of Difference-in-Discontinuities Pseudo-Effects for Placebo States



Notes: The figure reports the distribution of estimated pseudo-treatment effects from the difference-in-discontinuities specification (Column 3 in Table 1) for states that do not have a young child tax credit during the sample period using the PBT-optimal specification for the California analysis. The specification compares labor force participation among age-eligible versus age-ineligible mothers during the period with no work requirement (2022-2023) and the work requirement period (2019-2020). The distribution includes one estimated treatment effect for each placebo state (i.e., all states and the District of Columbia other than California, Colorado, Maryland, New Jersey, Oregon, and Vermont).

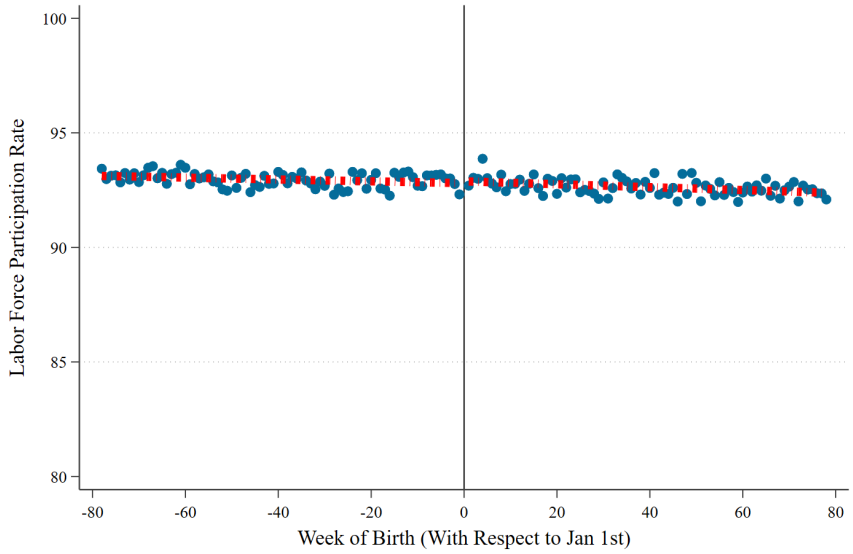
Figure A.11: Placebo-Based Tuning Analysis: Two-Year Specification



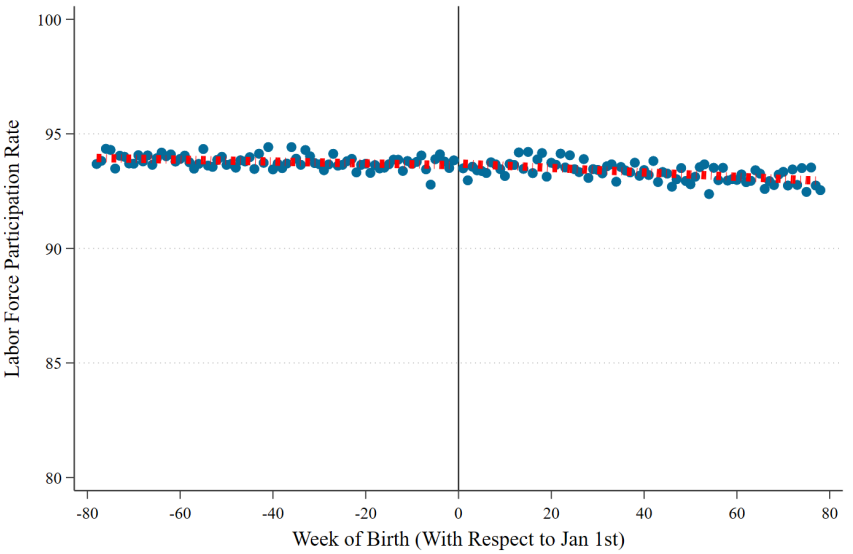
Notes: The figure replicates the analysis reported in Figure 2, but each placebo period spans two years rather than one. Each candidate specification includes a calendar year fixed effect. The placebo periods considered are: 2005 and 2006; 2006 and 2007; ...; 2017 and 2018.

Figure A.12: Labor Force Participation by Child's Date of Birth: Alternative Specification

(a) California, Work Requirement (2019-2020)

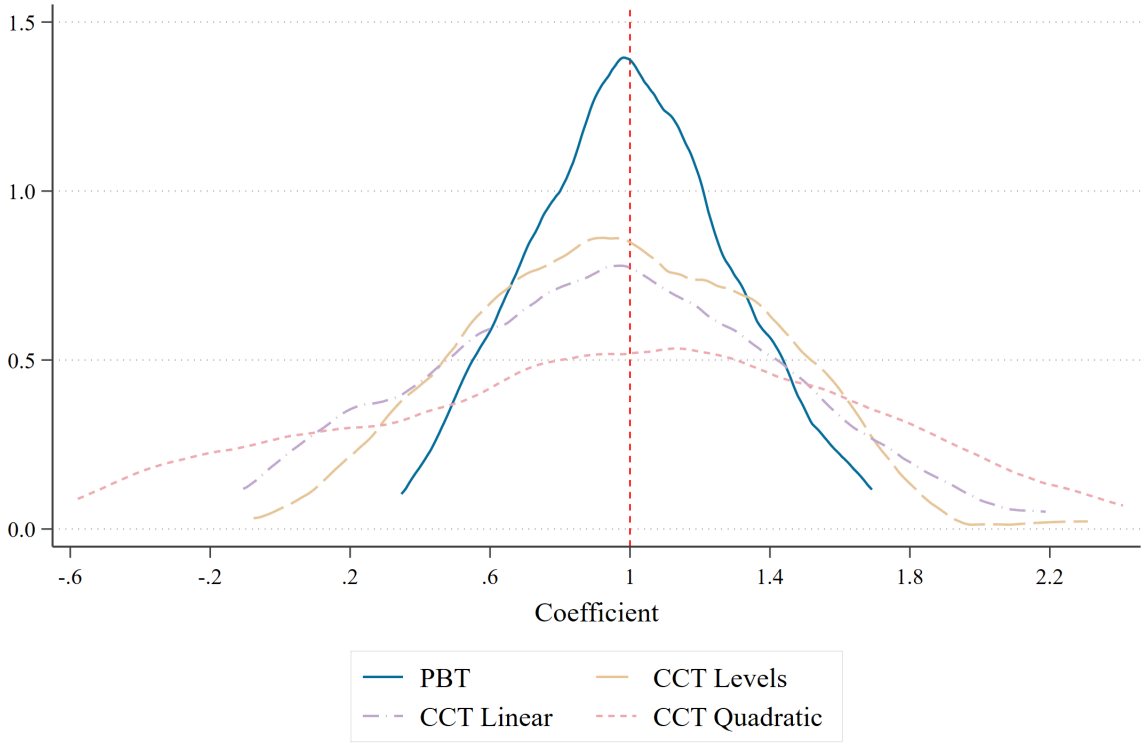


(b) California, No Work Requirement (2022-2023)



Notes: The figure reports mean labor force participation rates binned by child's week of birth for recently working California mothers. The sample consists of individuals who had positive wage or self-employment income reported on third-party information returns during the previous tax year and whose youngest child's sixth birthday falls within the 18-month window around the turn of the year (2019 or 2020 in Panel A; 2022 or 2023 in Panel B). The dashed lines correspond to the best linear fit, estimated separately for dates before versus after the turn of-the-year (normalized to zero).

Figure A.13: Simulation Analysis for PBT and Other Specification Selection Methods



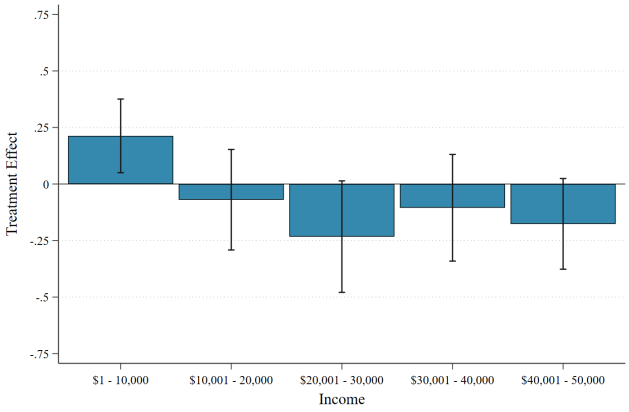
Notes: The figure reports results from a Monte Carlo simulation analysis comparing the performance of alternative approaches for selecting local polynomial order and bandwidth for regression discontinuity designs. PBT refers to the placebo-based tuning analysis described in Section 4. CCT refers to the optimal bandwidth selection method described in Calonico, Cattaneo and Titiunik (2014), implemented using the RDRobust Stata package with the package default (triangular) kernel; the levels, linear, and quadratic variants refer to the assumed local polynomial order. Each simulation iteration involves 14 placebo pre-periods ($t \in \{-13, -12, \ldots, 0\}$) and one post-period ($t = 1$). The running variable (X) ranges from -1 to 1 and is observed at 0.01 increments, so that there are 201 observations per period, indexed by i, each corresponding to a different value of X. The treatment ($T \in \{0, 1\}$) is applied to values of $X > 0$. The outcome (Y_{it}) is determined according to the following equation:

$$Y_{it} = \alpha_1 X_{it} + \alpha_2 X_{it}^2 + \alpha_3 X_{it}^3 + \alpha_4 X_{it}^4 + \beta_t T_{it} + \varepsilon_{it}$$

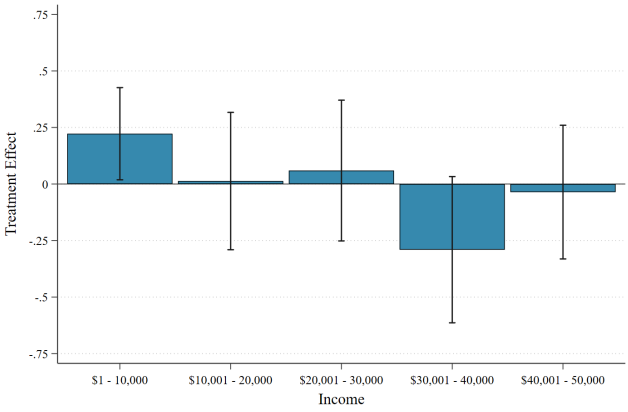
where β reflects the treatment effect and $\varepsilon_{it} \sim N(0, 1)$. The treatment effect is assumed to be zero for the placebo pre-periods, $\beta_t = 0$ for $t < 1$, and is equal to 1 for the post-period, $\beta_t = 1$ for $t = 1$. Each iteration of the simulation involves randomly drawing the vector of parameters that govern the relationship between the running variable and the outcome, $\alpha_k \sim U[-1, 1]$ for $k \in \{1, 2, 3, 4\}$. The plotted distributions are obtained as follows. For PBT, the optimal local polynomial order and bandwidth are determined based on the placebo pre-periods, following the procedure described in the notes for Figure 2, considering bandwidths from 0.1 to 1.0 in 0.1 step intervals. The estimated pseudo-treatment effect for each simulation is then obtained by applying the PBT-selected specification to the post-treatment year. The plotted distributions correspond to the distribution of estimated pseudo-treatment effects under the specified approach for selecting the empirical distribution.

Figure A.14: Intensive Margin Effect of California Credit Eligibility

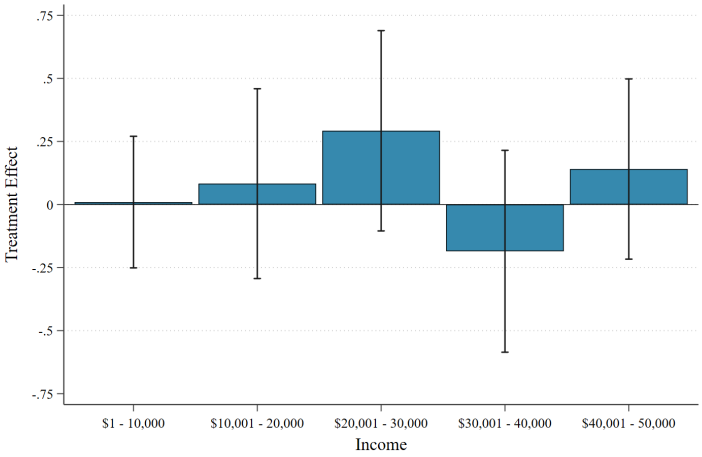
(a) California, Work Req. (2019-2020)



(b) California, No Work Req. (2022-2023)



(c) Difference-in-Discontinuities



Notes: The figure reports the estimated treatment effect of age-eligibility for the California credit on the likelihood that a taxpayer reports income within the specified income bin. The sample consists of individuals who had positive wage or self-employment income reported on third-party information returns during the previous tax year and whose youngest child's sixth birthday falls within the four-month window around the turn of the specified year. The sample is additionally limited to individuals who file a tax return reporting positive income. For each income bin, panels A, B, and C report specifications based on Columns 1, 2, and 3 (respectively) of Table 1. Treatment effects are reported in percentage points (0-100). Bars represent estimate 95% confidence intervals, derived from heteroskedasticity-robust standard errors.

Table A.1: Average State and Federal Tax Credit Eligibility by State

	California	Colorado	Maryland	New Jersey	Oregon	Vermont
Federal CTC	1,481.22	1,465.58	1,485.98	1,519.73	1,433.44	1,397.49
Federal EITC	3,459.23	3,419.79	3,448.84	3,499.16	3,329.57	3,247.63
State CTC	1,031.18	879.35	213.96	1,000.00	883.44	1,000.00
State EITC	1,834.89	1,299.52	1,724.42	1,399.67	299.66	1,234.10

Notes: The table reports average eligibility by state for federal EITC, federal CTC, state EITC, and state CTC for low-income tax filers with one child under age 6. Credit eligibility is calculated for tax year 2023 based on the federal tax return income and filing status reported in that year. Federal EITC, Federal CTC, and state EITC eligibility are calculated using NBER TaxSIM. State CTC eligibility is calculated manually based on the applicable statutory formula and state tax authority guidance.

Table A.2: Summary Statistics by Eligibility for California Credit

	(1)	(2)	(3)	(4)
	All	Age-Eligible (Jan-Apr)	Age-Ineligible (Sep-Dec)	p-value
Individual-Level				
Age	35.201	35.058	35.336	0.000
Any Income	1.000	1.000	1.000	.
Total Income	57,127	56,998	57,249	0.670
Self-Employed	0.119	0.119	0.118	0.550
Filed a Tax Return	0.934	0.935	0.933	0.034
Return-Level, if Filed				
Married	0.550	0.554	0.546	0.000
Num. Claimed Children	1.818	1.816	1.819	0.334
AGI	129,376	131,259	127,593	0.260
Claimed Federal EITC	0.374	0.370	0.377	0.000
Claimed Federal CTC	0.904	0.903	0.905	0.034
Observations	496,527	241,349	255,178	

Notes: The table reports lagged (prior-year) characteristics for recently working California mothers by age-eligibility for the YCTC. The sample consists of individuals who had positive wage or self-employment income reported on third-party information returns during the previous tax year and whose youngest child's sixth birthday falls within the four-month window around the turn of one of the following years: 2019, 2020, 2022, and 2023. Column 1 reports statistics for the full sample; Column 2 reports statistics for mothers of children whose birthday falls after the turn-of-the-year (age-eligible); Column 3 reports statistics for mothers of children whose birthday falls before the turn-of-the-year (age-ineligible); Column 4 reports the p-value for the test of equality between Columns 2 and 3. Individual-level characteristics are based on third-party information returns; return-level characteristics are based on tax return data and are only presented for individuals who filed a prior-year tax return.

Table A.3: California Credit Eligibility and Labor Force Participation: Robustness

	(1)	(2)	(3)
	Work Req. (2019-2020)	No Work Req. (2022-2023)	Diff-in-Disc [(2) - (1)]
Panel A: Has a Social Security Number			
Age-eligible	-0.005	-0.124	-0.119
	(0.106)	(0.102)	(0.147)
95% CI	[-0.21, 0.20]	[-0.32, 0.08]	[-0.41, 0.17]
Control Mean	93.8	94.4	94.1
Observations	210,661	205,769	416,430
Panel B: Claimed Child			
Age-eligible	-0.137	-0.092	0.045
	(0.101)	(0.097)	(0.140)
95% CI	[-0.33, 0.06]	[-0.28, 0.10]	[-0.23, 0.32]
Control Mean	93.4	94.1	93.7
Observations	246,066	238,559	484,625
	Work Req. DiD (2019-2021)	No Work Req. DiD (2022-2023)	Triple Difference [(2) - (1)]
Panel C: Triple Difference			
Age-eligible	-0.007	-0.065	-0.058
	(0.091)	(0.106)	(0.140)
95% CI	[-0.19, 0.17]	[-0.27, 0.14]	[-0.33, 0.22]
Control Mean	93.7	94.6	94.0
Observations	3,006,951	1,974,592	4,981,543

Notes: The table replicates the California YCTC analysis in Table 1 using alternative sample restrictions or time periods. Panel A restricts the sample to mothers assigned a social security number. Panel B restricts the sample to mothers who claimed the focal child on their prior-year tax return. Panel C is a triple-difference model including 2021, the year of the federal CTC expansion, in the analysis. The outcome in each column is an indicator for having positive wage or self-employment income reported on third-party information returns; units are percentage points (0-100).

Table A.4: Estimated Take-Up of the California Credit Among Federal Filers

	Number of Claimants			Total Credit Amount ($ in Mil)		
	(1)	(2)	(3)	(4)	(5)	(6)
	Claimed	Eligible	Takeup (%)	Claimed	Eligible	Takeup (%)
2019	428,857	517,406	82.9	389	465	83.7
2020	416,980	453,675	91.9	388	401	96.7
2019-2020	845,837	971,081	87.1	777	866	89.7

Notes: The table reports estimates of take-up of the California YCTC among federal tax filers. Take-up is estimated by comparing reported YCTC claiming statistics reported by the California Franchise Tax Board (FTB) to simulated YCTC eligibility for California federal tax filers claiming one or more children under the age of six. Calculations are conducted at the tax return level. Actual claim data regarding the number of YCTC claimants and the total dollars of YCTC claimed is derived from California Franchise Tax Board (2019) and the subsequent report for 2020. Simulated YCTC eligibility among federal filers is based on the YCTC credit formula and information reported on the federal return. For purposes of this exercise, we treat as eligible for the YCTC any federal filer satisfying the following criteria: had a California address; claimed at least one child below the age of six for either the federal CTC or EITC; reported Adjusted Gross Income under $30,000; reported positive wages or Schedule C profit; reported taxable interest and dividends below $3,828 for 2019 returns or $3,882 for 2020 returns; did not file as married filing separately; filed the federal return before the close of the calendar year following the tax year; and both the filer and child possessed a valid SSN. These criteria apply to the most recent tax filing. In the rare occasion that multiple taxpayers claimed the same child, the child was assigned to the taxpayer with higher income.

Table A.5: Yearly Estimates of California Credit Eligibility on Labor Force Participation

	(1)	(2)	(3)	(4)
	2019	2020	2021	2019-2021
Age-Eligibile x California	-0.069	-0.075	0.114	-0.101
	(0.153)	(0.156)	(0.163)	(0.141)
Age-Eligibile x California x Year				0.091
				(0.112)
Control Mean	94.1	93.7	93.2	93.7
Observations	1,077,818	1,095,161	1,082,917	3,255,896

Notes: The table reports the estimated effect of eligibility for the California YCTC separately for each year in which the credit was subject to a work requirement (2019-2021). The sample consists of recently working mothers whose youngest child's sixth birthday falls within the four-month window around the end of the specified year. The outcome in each column is an indicator for having positive wage or self-employment income reported on third-party information returns; units are percentage points (0-100). Columns 1 through 3 present the effect of YCTC age-eligibility by comparing labor force participation of California mothers of age-eligible versus age-ineligible children to mothers of same-aged children in different states in 2019 through 2021, respectively. Column 4 includes the samples for 2019 through 2021 adding an interaction between the indicator for YCTC age-eligibility (i.e., age-eligible and living in California) and a continuous year variable along with year by California fixed effects and year by age-eligibility fixed effects. The control mean corresponds to mothers of age-ineligible children. Parentheses report heteroskedasticity-robust standard errors.

Table A.6: Summary Statistics for Other States with Young Child Tax Credits

	(1)	(2)	(3)	(4)	(5)
	Colorado	Maryland	New Jersey	Oregon	Vermont
Individual-Level					
Age	34.341	35.023	35.628	34.527	34.912
Any Income	1.000	1.000	1.000	1.000	1.000
Total Income	50,357	62,064	64,764	50,929	48,823
Self-Employed	0.169	0.168	0.128	0.141	0.130
Filed a Tax Return	0.942	0.933	0.947	0.922	0.954
Return-Level, if Filed					
Married	0.670	0.558	0.603	0.628	0.670
Num. Claimed Children	2.078	1.937	1.896	1.902	1.976
AGI	119,170	122,765	145,269	111,190	108,217
Claimed Federal EITC	0.319	0.344	0.321	0.341	0.317
Claimed Federal CTC	0.914	0.928	0.907	0.922	0.924
Observations	74,873	138,541	441,346	91,310	13,746

Notes: The table reports lagged (prior-year) characteristics for recently working mothers in states other than California with state-level young child tax credits. The sample consists of individuals who had positive wage or self-employment income reported on third-party information returns during the previous tax year and whose child's sixth birthday falls within the PBT-specified bandwidth around the turn of the year for the years in which the applicable state credit was available. Individual-level characteristics are based on third-party information returns; return-level characteristics are based on tax return data and are only presented for individuals who filed a prior-year tax return.

Table A.7: Yearly Estimates of Colorado Credit Eligibility on Labor Force Participation

	(1)	(2)
	2022	2023
Age-eligible	0.185	-0.121
	(0.257)	(0.262)
95% CI	[-0.32, 0.69]	[-0.63, 0.39]
Control Mean	93.3	93.1
Observations	37,382	37,491

Notes: The table reports the estimated effect of eligibility for the Colorado CTC separately for each year that the credit was in effect (2022 and 2023). The sample consists of recently working mothers with a child whose sixth birthday falls within the five-month window around the end of the specified year. The outcome in each column is an indicator for having positive wage or self-employment income reported on third-party information returns; units are percentage points (0-100).

Table A.8: Estimated Pseudo Treatment Effects from the California Pre-Period

	(1)	(2)	(3)	(4)	(5)
	Leave-One-Out	CCT Levels	CCT Linear	CCT Quadratic	Commonly Used Specification
Age-eligible	-0.012	-0.088	-0.014	-0.206	0.440
	(0.216)	(0.232)	(0.344)	(0.476)	(1.050)
Control Mean	92.3	92.2	92.3	92.3	92.1

Notes: The table reports the mean and standard deviation of the estimated effect of YCTC age-eligibility on labor force participation for each year from 2005 through 2018 (prior to the YCTC's introduction). Units are percentage points (0-100). The columns report alternative approaches for estimating the yearly treatment effects. In Column 1, a leave-one-out procedure is employed, in which each year's specification is selected by applying the placebo-based tuning analysis to the other 13 pre-period years. In Columns 2-4, the specification is selected using the Calonico, Cattaneo and Titiunik (2014) optimal bandwidth method, implemented using the RDRobust Stata package, imposing a local polynomial in child date of birth with the specified order and the package default (triangular) kernel. In Column 5, the specification is a regression discontinuity design with a one-month bandwidth, a local linear polynomial with uniform bandwidth, and an 8-day donut around the turn-of-the-year.

Table A.9: Other State Pseudo Effects During the Pre-Period

	(1)	(2)	(3)	(4)	(5)
	Leave-One-Out	CCT Levels	CCT Linear	CCT Quadratic	Commonly Used Specification
Colorado	-0.018	0.147	0.328	-0.044	-0.486
	(0.193)	(0.394)	(0.688)	(0.987)	(3.514)
New Jersey	0.028	0.014	0.101	0.034	0.980
	(0.138)	(0.329)	(0.413)	(0.572)	(1.932)
Maryland	-0.033	0.132	0.170	0.146	-0.707
	(0.196)	(0.327)	(0.385)	(0.723)	(2.323)
Oregon	-0.035	-0.159	-0.152	-0.389	1.210
	(0.362)	(0.379)	(0.731)	(1.017)	(3.667)
Vermont	-0.276	0.474	0.336	0.211	0.427
	(0.552)	(1.805)	(2.239)	(2.983)	(8.546)

Notes: The table replicates the analysis in Table A.8 for states other than California.

Table A.10: Placebo Tests: California Credit Eligibility and Labor Force Participation

	(1)	(2)	(3)
	Work Req. (2019-2020)	No Work Req. (2022-2023)	Diff-in-Disc [(2) - (1)]
Panel A: All States Sample			
Age-eligible	-0.036	-0.053	-0.017
	(0.036)	(0.034)	(0.049)
95% CI	[-0.11, 0.03]	[-0.12, 0.01]	[-0.11, 0.08]
Control Mean	94.0	94.7	94.4
Observations	1,757,313	1,733,838	3,491,151
Panel B: Younger Child Sample			
Age-eligible	0.029	0.084	0.055
	(0.160)	(0.160)	(0.226)
95% CI	[-0.28, 0.34]	[-0.23, 0.40]	[-0.39, 0.50]
Control Mean	89.7	90.8	90.2
Observations	145,447	131,776	277,223

Notes: Panel A replicates Table 1 but for individuals living in placebo states that did not offer a young child tax credit during our sample period (i.e., all states and the District of Columbia other than California, Colorado, Maryland, New Jersey, Oregon, and Vermont). Panel B replicates Table 1 for California mothers with a child that turns six within the four months around the turn-of-the-year *in addition to* a younger child who would continue to qualify the mother for the California YCTC during the outcome year.

Table A.11: Investigating Time Variation in the Effect of California Credit Eligibility

	(1)	(2)	(3)
	2022-23 vs 2019	2023 vs 2019	2020 vs 2019
Age-eligible	0.024	-0.039	0.049
	(0.176)	(0.204)	(0.207)
95% CI	[-0.32, 0.37]	[-0.44, 0.36]	[-0.36, 0.45]
Control Mean	93.4	93.1	92.8
Observations	370,060	248,043	251,699

Notes: The table compares labor force participation of recently working California mothers with children born around the turn of the year for various combinations of outcome years. The outcome in each column is an indicator for having positive wage or self-employment income reported on third-party information returns; units are percentage points (0-100). Column 1 reports the difference-in-discontinuities estimate from equation 3 excluding 2020. Column 2 reports the same specification but excludes both 2020 and 2022. Column 3 compares mothers of age-eligible versus age-ineligible children in California across the two years within the YCTC work requirement period.

Table A.12: PBT-Optimal Linear RD Specification: California

	(1)	(2)	(3)
	Work Req. (2019-2020)	No Work Req. (2022-2023)	Diff-in-Disc [(2) - (1)]
Age-eligible	0.072	0.069	-0.003
	(0.104)	(0.100)	(0.144)
95% CI	[-0.13, 0.27]	[-0.13, 0.27]	[-0.29, 0.28]
Control Mean	93.0	93.8	93.4
Observations	988,326	958,470	1,946,796

Notes: The table replicates Table 1 under an alternative regression discontinuity specification with a local linear polynomial in child's date-of-birth and a bandwidth of 18 months. The specification uses a uniform kernel and does not exclude a donut of birthdates near the turn-of-the-year.

Table A.13: CTC Eligibility and Labor Force Participation by RD Model

	Work Requirement		No Work Requirement				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	California	Colorado	California	Maryland	New Jersey	Oregon	Vermont
Panel A: Commonly Used RD							
Coefficient	-0.285	2.447	-0.172	0.545	0.370	-1.770	4.797
	(0.738)	(1.464)	(0.711)	(1.782)	(1.026)	(2.636)	(4.660)
95% CI	[-1.73, 1.16]	[-0.42, 5.32]	[-1.57, 1.22]	[-2.95, 4.04]	[-1.64, 2.38]	[-6.94, 3.40]	[-4.35, 13.94]
Panel B: CCT Method							
Coefficient	0.110	-0.023	-0.444	-0.350	0.325	-0.063	-1.236
	(0.401)	(1.022)	(0.457)	(0.573)	(0.630)	(0.761)	(3.120)
95% CI	[-0.67, 0.90]	[-2.03, 1.98]	[-1.34, 0.45]	[-1.47, 0.77]	[-0.91, 1.56]	[-1.55, 1.43]	[-7.35, 4.88]

Notes: The table estimates the effect of state CTC eligibility using two alternative regression discontinuity specifications. Panel A uses a local linear polynomial in child's date-of-birth, a bandwidth of one month, a uniform kernel, and a donut exclusion of 8 days around the turn-of-the-year. Panel B uses the optimal bandwidth selection method proposed in Calonico, Cattaneo and Titiunik (2014), implemented using the RDRobust Stata package, with a linear polynomial in child's date-of-birth, a triangular kernel, and no donut exclusion. The optimal specification is selected separately for each tax year and estimates are aggregated across years within the same state and policy period. Columns 1 and 2 correspond to state-years for which the applicable credit is conditioned on work; Columns 3-7 correspond to state-years for which the applicable credit is available to non-workers. Column 1 includes California years for 2019 and 2020; Column 3 includes California years for 2022 and 2023. Other aspects of the analysis follow Tables 1 and 2.

Table A.14: Alternative Sample Based on Working Prior to Credit Adoption

	(1)	(2)	(3)	(4)
	California	Colorado	New Jersey	Vermont
Panel A: First Year After Adoption				
Coefficient	-0.111	0.256	-0.003	-0.202
	(0.144)	(0.256)	(0.179)	(0.555)
95% CI	[-0.39, 0.17]	[-0.25, 0.76]	[-0.35, 0.35]	[-1.29, 0.89]
Control Mean	92.9	93.3	95.5	94.4
Observations	127,137	37,577	219,546	6,956
Panel B: Second Year After Adoption				
Coefficient	-0.043	0.051	0.137	-0.130
	(0.141)	(0.246)	(0.169)	(0.527)
95% CI	[-0.32, 0.23]	[-0.43, 0.53]	[-0.20, 0.47]	[-1.16, 0.90]
Control Mean	93.8	94.5	96.0	95.5
Observations	118,254	34,342	206,658	6,326

Notes: The Table reports the estimated effect of eligibility for the specified state CTC for the sample of mothers who were working in the year prior to the adoption of the credit in the specified state. The analysis is limited to states for which we observe at least two years of outcome data in years following the credit's adoption. For California, the adoption considered by the analysis is the initial adoption of the credit, not its subsequent expansion. Panels A and B respectively report effects for the first and second years following the credit's adoption. The outcome in each column is an indicator for having positive wage or self-employment income reported on third-party information returns; units are percentage points (0-100). Parentheses report heteroskedasticity-robust standard errors.

Table A.15: CTC Eligibility and Labor Force Participation by Income

	Work Requirement		No Work Requirement				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	California	Colorado	California	Maryland	New Jersey	Oregon	Vermont
Panel A: Lower Income (< $30,000)							
Coefficient	-0.203	0.398	-0.234	-0.857	0.287	-0.491	-1.286
	(0.219)	(0.460)	(0.242)	(0.601)	(0.339)	(0.822)	(1.149)
95% CI	[-0.63, 0.23]	[-0.50, 1.30]	[-0.71, 0.24]	[-2.03, 0.32]	[-0.38, 0.95]	[-2.10, 1.12]	[-3.54, 0.97]
Control Mean	88.4	88.2	88.4	89.6	90.7	87.5	88.0
Observations	86,036	19,419	70,890	40,558	116,461	25,992	3,357
Panel B: Higher Income (≥ $30,000)							
Coefficient	-0.075	-0.116	-0.099	0.270	-0.174	0.089	-0.174
	(0.106)	(0.186)	(0.098)	(0.222)	(0.124)	(0.321)	(0.352)
95% CI	[-0.28, 0.13]	[-0.48, 0.25]	[-0.29, 0.09]	[-0.16, 0.71]	[-0.42, 0.07]	[-0.54, 0.72]	[-0.86, 0.51]
Control Mean	95.1	95.0	95.7	96.9	96.8	95.7	96.8
Observations	165,663	55,454	173,938	97,983	324,885	65,318	10,389

Notes: The table replicates Table 1 and 2 separately for taxpayers with prior-year reported AGI below versus above $30,000. Taxpayers who did not file during the prior-year are treated as having reported an AGI of $0.

Table A.16: CTC Eligibility and Labor Force Participation by Marital Status

	Work Requirement		No Work Requirement				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	California	Colorado	California	Maryland	New Jersey	Oregon	Vermont
Panel A: Married							
Coefficient	0.046	-0.075	-0.025	0.418	-0.070	-0.014	-0.070
	(0.142)	(0.227)	(0.137)	(0.303)	(0.162)	(0.408)	(0.429)
95% CI	[-0.23, 0.32]	[-0.52, 0.37]	[-0.29, 0.24]	[-0.18, 1.01]	[-0.39, 0.25]	[-0.81, 0.79]	[-0.91, 0.77]
Control Mean	93.0	93.5	93.7	95.5	95.6	94.2	95.8
Observations	129,239	47,221	125,786	72,151	251,963	52,890	8,785
Panel B: Unmarried							
Coefficient	-0.145	0.090	-0.204	-0.412	0.115	0.387	-1.823*
	(0.142)	(0.295)	(0.135)	(0.355)	(0.198)	(0.501)	(0.730)
95% CI	[-0.42, 0.13]	[-0.49, 0.67]	[-0.47, 0.06]	[-1.11, 0.28]	[-0.27, 0.50]	[-0.60, 1.37]	[-3.26, -0.39]
Control Mean	94.4	94.6	95.2	95.3	95.9	95.0	94.8
Observations	106,635	23,273	102,130	57,090	166,140	31,324	4,329

Notes: The table replicates Table 1 and 2 separately for married versus unmarried taxpayers. Marital status is measured based on taxpayers' prior-year filing status. Taxpayers who filed as married filing jointly or married filing separately are treated as married. Taxpayers who filed with a different filing status are treated as unmarried. Taxpayers who did not file a prior-year tax return are excluded from this analysis. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A.17: CTC Eligibility and Labor Force Participation by Marital Status and Income

	Work Requirement		No Work Requirement				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	California	Colorado	California	Maryland	New Jersey	Oregon	Vermont
Panel A: Lower Income and Married							
Coefficient	0.862	1.246	-0.566	0.144	2.369	-0.196	0.223
	(0.757)	(1.365)	(0.895)	(2.382)	(1.295)	(2.677)	(3.167)
95% CI	[-0.62, 2.35]	[-1.43, 3.92]	[-2.32, 1.19]	[-4.53, 4.82]	[-0.17, 4.91]	[-5.45, 5.05]	[-6.00, 6.45]
Control Mean	83.2	84.3	83.8	86.3	85.5	84.8	87.6
Observations	9,557	2,741	6,902	3,314	11,016	2,965	433
Panel B: Higher Income and Married							
Coefficient	-0.027	-0.163	-0.013	0.424	-0.200	0.018	-0.044
	(0.140)	(0.225)	(0.134)	(0.295)	(0.157)	(0.400)	(0.419)
95% CI	[-0.30, 0.25]	[-0.60, 0.28]	[-0.28, 0.25]	[-0.15, 1.00]	[-0.51, 0.11]	[-0.77, 0.80]	[-0.87, 0.78]
Control Mean	93.8	94.1	94.3	96.0	96.1	94.7	96.2
Observations	119,682	44,480	118,884	68,837	240,947	49,925	8,352
Panel C: Lower Income and Unmarried							
Coefficient	-0.115	0.049	-0.191	-0.752	0.318	0.461	-2.466
	(0.231)	(0.515)	(0.265)	(0.665)	(0.365)	(0.901)	(1.273)
95% CI	[-0.57, 0.34]	[-0.96, 1.06]	[-0.71, 0.33]	[-2.06, 0.55]	[-0.40, 1.03]	[-1.31, 2.23]	[-4.96, 0.03]
Control Mean	91.2	91.0	91.0	91.2	92.5	91.0	90.9
Observations	60,654	12,299	47,076	27,944	82,202	15,931	2,292
Panel D: Higher Income and Unmarried							
Coefficient	0.010	0.190	-0.148	0.001	-0.026	0.251	-0.558
	(0.112)	(0.217)	(0.100)	(0.240)	(0.141)	(0.368)	(0.509)
95% CI	[-0.21, 0.23]	[-0.24, 0.62]	[-0.34, 0.05]	[-0.47, 0.47]	[-0.30, 0.25]	[-0.47, 0.97]	[-1.56, 0.44]
Control Mean	98.5	98.6	98.7	98.9	99.0	98.9	99.0
Observations	45,981	10,974	55,054	29,146	83,938	15,393	2,037

Notes: The table replicates Table 1 and 2 separately by prior-year income (reported AGI below versus above $30,000) and marital status (married versus unmarried). Taxpayers who did not file during the prior-year are treated as having reported an AGI of $0. Marital status is measured based on taxpayers' prior-year filing status. Taxpayers who filed as married filing jointly or married filing separately are treated as married. Taxpayers who filed with a different filing status are treated as unmarried. Taxpayers who did not file a prior-year tax return are excluded from this analysis.

Table A.18: CTC Eligibility and Labor Force Participation by Self Employment Status

	Work Requirement		No Work Requirement				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	California	Colorado	California	Maryland	New Jersey	Oregon	Vermont
Panel A: Has Self-Employment Income							
Coefficient	-0.124	-0.806	-0.178	-0.039	-0.070	-1.554	-0.874
	(0.440)	(0.580)	(0.359)	(0.797)	(0.162)	(1.151)	(1.497)
95% CI	[-0.99, 0.74]	[-1.94, 0.33]	[-0.88, 0.52]	[-1.60, 1.52]	[-0.39, 0.25]	[-3.81, 0.70]	[-3.81, 2.06]
Control Mean	85.0	88.3	88.3	89.0	89.9	88.2	89.2
Observations	26,430	12,670	32,425	23,270	251,963	12,885	1,788
Panel B: Does Not Have Self-Employment Income							
Coefficient	-0.100	0.188	-0.103	-0.083	0.020	0.186	-0.499
	(0.103)	(0.185)	(0.100)	(0.232)	(0.126)	(0.333)	(0.390)
95% CI	[-0.30, 0.10]	[-0.18, 0.55]	[-0.30, 0.09]	[-0.54, 0.37]	[-0.23, 0.27]	[-0.47, 0.84]	[-1.26, 0.27]
Control Mean	93.7	94.2	94.4	96.0	96.0	94.3	95.5
Observations	225,269	62,203	212,403	115,271	385,053	78,425	11,958

Notes: The table replicates Table 1 and 2 separately for taxpayers with and without prior-year self-employment income. Taxpayers are treated as having self-employment income if they have positive reported values on an information return 1099-MISC or 1099-NEC.

Table A.19: CTC Eligibility and Labor Force Participation by Urbanicity

	Work Requirement		No Work Requirement				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	California	Colorado	California	Maryland	New Jersey	Oregon	Vermont
Panel A: Lived in Urban Zip Code							
Coefficient	-0.184	-0.045	-0.109	-0.106	-0.031	0.047	-0.611
	(0.106)	(0.196)	(0.102)	(0.246)	(0.129)	(0.376)	(0.673)
95% CI	[-0.39, 0.02]	[-0.43, 0.34]	[-0.31, 0.09]	[-0.59, 0.38]	[-0.28, 0.22]	[-0.69, 0.78]	[-1.93, 0.71]
Control Mean	92.9	93.4	93.7	94.8	95.3	93.6	95.3
Observations	234,896	64,871	228,564	128,896	432,806	68,970	4,182
Panel B: Did Not Live in Urban Zip Code							
Coefficient	0.829	0.615	-0.273	0.561	0.118	-0.456	-0.514
	(0.434)	(0.530)	(0.422)	(0.876)	(1.041)	(0.686)	(0.482)
95% CI	[-0.02, 1.68]	[-0.42, 1.65]	[-1.10, 0.55]	[-1.16, 2.28]	[-1.92, 2.16]	[-1.80, 0.89]	[-1.46, 0.43]
Control Mean	91.3	92.2	92.5	94.6	94.0	92.9	94.4
Observations	16,211	9,884	15,797	9,626	8,503	22,324	9,558

Notes: The table replicates Table 1 and 2 separately for taxpayers living in an urban zip code (Rural-Urban Commuting Area (RUCA) less than 3) versus those that do not.

Table A.20: CTC Eligibility and Labor Force Participation by College Attendance

	Work Requirement		No Work Requirement				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	California	Colorado	California	Maryland	New Jersey	Oregon	Vermont
Panel A: Any College							
Coefficient	-0.252*	-0.088	-0.002	0.151	-0.055	-0.347	-0.153
	(0.113)	(0.192)	(0.106)	(0.229)	(0.132)	(0.360)	(0.400)
95% CI	[-0.47, -0.03]	[-0.46, 0.29]	[-0.21, 0.21]	[-0.30, 0.60]	[-0.31, 0.20]	[-1.05, 0.36]	[-0.94, 0.63]
Control Mean	94.4	94.4	95.0	96.4	96.4	94.4	95.5
Observations	170,231	57,872	169,092	102,821	319,351	67,223	10,890
Panel B: No College							
Coefficient	0.048	0.304	-0.482*	-0.854	-0.039	0.718	-2.067
	(0.214)	(0.471)	(0.216)	(0.634)	(0.308)	(0.744)	(1.104)
95% CI	[-0.37, 0.47]	[-0.62, 1.23]	[-0.91, -0.06]	[-2.10, 0.39]	[-0.64, 0.56]	[-0.74, 2.18]	[-4.23, 0.10]
Control Mean	89.6	89.4	90.5	90.0	92.4	90.8	91.5
Observations	81,468	17,001	75,736	35,720	121,995	24,087	2,856

Notes: The table replicates Table 1 and 2 separately for taxpayers based on previous college attendance. A taxpayer is treated as having attended college if the taxpayer received a Form 1098-T for any prior year.

Table A.21: CTC Eligibility and Labor Force Participation by Presence of Additional Younger Child

	Work Requirement	No Work Requirement			
	(1)	(2)	(3)	(4)	(5)
	Colorado	Maryland	New Jersey	Oregon	Vermont
Panel A: Additional Younger Child					
Coefficient	0.049	-0.013	-0.050	0.182	-1.439*
	(0.329)	(0.453)	(0.260)	(0.698)	(0.729)
95% CI	[-0.59, 0.69]	[-0.90, 0.87]	[-0.56, 0.46]	[-1.19, 1.55]	[-2.87, -0.01]
Control Mean	91.6	93.6	94.0	91.0	93.7
Observations	28,391	41,804	122,523	23,919	4,967
Panel B: No Additional Younger Child					
Coefficient	-0.022	-0.072	0.023	-0.046	-0.111
	(0.217)	(0.276)	(0.146)	(0.370)	(0.458)
95% CI	[-0.45, 0.40]	[-0.61, 0.47]	[-0.26, 0.31]	[-0.77, 0.68]	[-1.01, 0.79]
Control Mean	94.2	95.2	95.6	94.1	95.2
Observations	46,482	96,737	318,823	67,391	8,779

Notes: The table replicates Table 2 separately for taxpayers with at least one additional child younger than the focal child in the household. These analyses exclude California as California's child tax credit is per return rather than per child.

Table A.22: CTC Eligibility and Labor Force Participation for Low-Income, Unmarried Mothers without Education

	Work Requirement		No Work Requirement				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	California	Colorado	California	Maryland	New Jersey	Oregon	Vermont
Coefficient	-0.156	0.642	-0.415	-1.381	-0.028	1.324	-6.015**
	(0.370)	(0.848)	(0.423)	(1.084)	(0.593)	(1.356)	(2.009)
95% CI	[-0.88, 0.57]	[-1.02, 2.30]	[-1.24, 0.41]	[-3.51, 0.74]	[-1.19, 1.13]	[-1.33, 3.98]	[-9.96, -2.07]
Control Mean	89.9	89.3	89.7	88.5	91.0	90.7	91.6
Observations	26,583	5,174	21,129	13,179	37,227	7,238	1,006

Notes: The table replicates Table 1 and 2 for unmarried mothers with no college education whose prior-year AGI was below $30,000. Taxpayers who did not file during the prior-year are treated as having reported an AGI of $0. Marital status is measured based on taxpayers' prior-year filing status. Taxpayers who filed as married filing jointly or married filing separately are treated as married; taxpayers who filed with a different filing status are treated as unmarried. A taxpayer is treated as having attended college if the taxpayer received a Form 1098-T for any prior year. Taxpayers who did not file a prior-year tax return are excluded from this analysis.

Table A.23: Medicaid Sample: Summary Statistics

	(1)	(2)	(3)	(4)	(5)	(6)
	California	Colorado	Maryland	New Jersey	Oregon	Vermont
Individual-Level						
Age	33.605	32.233	33.042	33.481	32.929	33.097
Any Income	0.669	0.688	0.734	0.698	0.679	0.695
Total Income	16,394	17,195	22,819	18,482	18,650	18,635
Self-Employed	0.083	0.128	0.156	0.109	0.110	0.100
Filed a Tax Return	0.839	0.857	0.836	0.865	0.791	0.838
Return-Level, if Filed						
Married	0.439	0.507	0.353	0.397	0.518	0.513
Num. Claimed Children	1.932	2.260	2.025	2.034	2.052	2.102
AGI	38,516	44,581	42,027	39,212	48,757	46,773
Claimed Federal EITC	0.594	0.586	0.621	0.626	0.573	0.569
Claimed Federal CTC	0.921	0.927	0.934	0.938	0.921	0.915
Observations	374,647	47,586	91,951	255,434	73,181	10,143

Notes: The table reports lagged (prior-year) characteristics for mothers who were enrolled in Medicaid during the prior-year in states with young child tax credits. The sample for each state consists of mothers whose child's sixth birthday falls within the PBT-specified bandwidth around the turn-of-the-year for the years in which the applicable state credit was available; PBT-selected bandwidths by state for the Medicaid sample are reported in Table 2. Individual-level characteristics are based on third-party information returns; return-level characteristics are based on tax return data and are only presented for individuals who filed a prior-year tax return.

Table A.24: California Credit Eligibility and Labor Force Participation: Alternative Samples

	(1)	(2)	(3)
	Work Req. (2019-2020)	No Work Req. (2022-2023)	Diff-in-Disc [(2) - (1)]
Panel A: Medicaid Sample			
Age-eligible	0.031	0.265	0.234
	(0.216)	(0.218)	(0.306)
95% CI	[-0.39, 0.45]	[-0.16, 0.69]	[-0.37, 0.83]
Control Mean	66.2	68.4	67.3
Observations	192,234	182,413	374,647
Panel B: Census Sample			
Age-eligible	0.072	0.268*	0.196
	(0.145)	(0.128)	(0.194)
95% CI	[-0.21, 0.36]	[0.02, 0.52]	[-0.18, 0.58]
Control Mean	70.5	68.6	69.4
Observations	400,000	530,000	920,000

Notes: The table replicates the California YCTC analysis in Table 1 for the Medicaid sample (Panel A) and Census sample (Panel B). The Medicaid sample includes mothers of children enrolled in Medicaid in the state of California at any point during the prior year. The Census sample is comprised of birth mothers of children born in California; birth parents are identified from the Census Household Composition Key (CHCK) and birth location is according to the Social Security Administration (SSA) Numident file. For the Medicaid sample, the outcome in each column is an indicator for having positive wage or self-employment income reported on third-party information returns. For the Census sample, the outcome is an indicator for positive wage income reported on a W-2 form; all counts and estimates are rounded per Census's disclosure rules governing administrative records. Units are percentage points (0-100). Parentheses report heteroskedasticity-robust standard errors. The Census Bureau has reviewed results from the Census sample to ensure ensure appropriate access, use, and disclosure avoidance protection of the confidential source data used to produce this product (Data Management System (DMS) Number: P-7503840, Disclosure Review Board (DRB) approval number: CBDRB-FY24-SEHSD003-066.) * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A.25: Other State Credit Eligibility and Labor Force Participation: Alternative Sample

	(1)	(2)	(3)	(4)	(5)
	Colorado	Maryland	New Jersey	Oregon	Vermont
Age-eligible	0.770	-0.763	-0.040	0.205	-0.843
	(0.417)	(0.574)	(0.348)	(0.674)	(0.899)
95% CI	[-0.05, 1.59]	[-1.89, 0.36]	[-0.72, 0.64]	[-1.12, 1.53]	[-2.61, 0.92]
Control Mean	70.5	73.3	71.9	69.3	71.6
Observations	47,586	91,951	255,434	73,181	10,143

Notes: The table replicates the analysis from Table 2 for the Medicaid sample. The Medicaid sample is comprised of mothers of children enrolled in Medicaid in the specified state at any point during the prior year.

Table A.26: California Credit Eligibility and Income Reporting

	(1)	(2)	(3)
	Work Req. (2019-2020)	No Work Req. (2022)	Diff-in-Disc [(2) - (1)]
Panel A: Reported Wages			
Age-eligible	0.076	0.362*	0.287
	(0.124)	(0.185)	(0.222)
95% CI	[-0.17, 0.32]	[0.00, 0.72]	[-0.15, 0.72]
Control Mean	89.2	88.0	88.8
Observations	251,699	122,017	373,716
Panel B: Reported Schedule SE Income			
Age-eligible	0.024	-0.091	-0.115
	(0.136)	(0.203)	(0.244)
95% CI	[-0.24, 0.29]	[-0.49, 0.31]	[-0.59, 0.36]
Control Mean	13.4	14.8	13.8
Observations	251,699	122,017	373,716
Panel C: Reported Earned Income			
Age-eligible	-0.006	0.303	0.309
	(0.110)	(0.163)	(0.196)
95% CI	[-0.22, 0.21]	[-0.02, 0.62]	[-0.08, 0.69]
Control Mean	91.7	91.0	91.5
Observations	251,699	122,017	373,716

Notes: The table replicates the analysis in Table 1 for outcomes based on the income that taxpayers report on their federal tax returns. The outcomes considered in each panel are: an indicator for reporting positive wage income (Panel A); an indicator for reporting positive self-employment income (Panel B); and an indicator for reporting positive earned income (defined as the sum of wage and self-employment income) (Panel C). Units are percentage points (0-100). Each outcome takes on a value of zero if the individual does not file a tax return. Self-employment income is measured as the sum of net profits reported on each Schedule SE that the taxpayer files. The analysis excludes filing outcomes for tax year 2023, for which filing data was not available to us at the time of our analysis. Tax returns were excluded from the analysis if they were filed after the end of the calendar year following the tax year covered by the return. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A.27: Other State Credit Eligibility and Income Reporting

	(1)	(2)	(3)
	Colorado	New Jersey	Vermont
Panel A: Reported Wages			
Age-eligible	0.194	-0.077	0.074
	(0.316)	(0.240)	(0.659)
95% CI	[-0.43, 0.81]	[-0.55, 0.39]	[-1.22, 1.37]
Control Mean	89.5	91.1	91.8
Observations	37,382	217,986	6,937
Panel B: Reported Schedule SE Income			
Age-eligible	-0.522	0.062	-0.151
	(0.409)	(0.322)	(0.951)
95% CI	[-1.32, 0.28]	[-0.57, 0.69]	[-2.02, 1.71]
Control Mean	19.6	17.6	19.6
Observations	37,382	217,986	6,937
Panel C: Reported Earned Income			
Age-eligible	0.208	0.005	0.062
	(0.275)	(0.206)	(0.578)
95% CI	[-0.33, 0.75]	[-0.40, 0.41]	[-1.07, 1.20]
Control Mean	92.2	93.6	93.8
Observations	37,382	217,986	6,937

Notes: The table replicates the analysis in Table 2 for outcomes based on the income that taxpayers report on their federal tax returns. The analysis is limited to Colorado, New Jersey, and Vermont, which provided a state-level CTC for tax year 2022 – the most recent year for which reported income data is available to us at the time of our analysis. For the same reason, the analysis excludes reporting outcomes for tax year 2023. The outcomes considered in each panel are: an indicator for reporting positive wage income (Panel A); an indicator for reporting positive self-employment income (Panel B); and an indicator for reporting positive earned income (defined as the sum of wage and self-employment income) (Panel C). Units are percentage points (0-100). Each outcome takes on a value of zero if the individual does not file a tax return. Self-employment income is measured as the sum of net profits reported on each Schedule SE that the taxpayer files. Tax returns were excluded from the analysis if they were filed after the end of the calendar year following the tax year covered by the return.

Table A.28: California Credit Eligibility and Tax Filing

	(1)	(2)	(3)
	Work Req. (2019-2020)	No Work Req. (2022)	Diff-in-Disc [(2) - (1)]
Age-eligible	0.014	0.365*	0.351
	(0.100)	(0.160)	(0.188)
95% CI	[-0.18, 0.21]	[0.05, 0.68]	[-0.02, 0.72]
Control Mean	93.3	91.3	92.6
Observations	251,699	122,017	373,716

Notes: The table replicates the analysis in Table 1 for the outcome of filing a federal tax return. The analysis excludes filing outcomes for tax year 2023, for which filing data was not available to us at the time of our analysis. Units are percentage points (0-100). Tax returns were excluded from the analysis if they were filed after the end of the calendar year following the tax year covered by the return. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A.29: Other State Credit Eligibility and Tax Filing

	(1)	(2)	(3)
	Colorado	New Jersey	Vermont
Age-eligible	0.220	-0.002	-0.157
	(0.268)	(0.202)	(0.564)
95% CI	[-0.31, 0.75]	[-0.40, 0.39]	[-1.26, 0.95]
Control Mean	92.7	93.9	94.2
Observations	37,382	217,986	6,937

Notes: The table replicates the analysis in Table 2 for the outcome of filing a federal tax return. The analysis is limited to Colorado, New Jersey, and Vermont, which provided a state-level CTC for tax year 2022 – the most recent year for which tax filing data is available to us at the time of our analysis. For the same reason, the analysis excludes filing outcomes for tax year 2023. Units are percentage points (0-100). Tax returns were excluded from the analysis if they were filed after the end of the calendar year following the tax year covered by the return.

Table A.30: California Credit Eligibility and Small-Dollar Income Reporting

	(1)	(2)	(3)
	Work Req. (2019-2020)	No Work Req. (2022)	Diff-in-Disc [(2) - (1)]
Panel A: Earned Income between 1 and 1,000			
Age-eligible	0.044	0.022	-0.022
	(0.024)	(0.024)	(0.034)
95% CI	[-0.00, 0.09]	[-0.02, 0.07]	[-0.09, 0.04]
Control Mean	0.3	0.2	0.3
Observations	251,699	122,017	373,716
Panel B: Earned Income between 1 and 5,000			
Age-eligible	0.178**	0.137*	-0.041
	(0.061)	(0.066)	(0.090)
95% CI	[0.06, 0.30]	[0.01, 0.27]	[-0.22, 0.13]
Control Mean	2.4	1.3	2.0
Observations	251,699	122,017	373,716
Panel C: Earned Income between 1 and 10,000			
Age-eligible	0.152	0.327**	0.175
	(0.096)	(0.110)	(0.146)
95% CI	[-0.04, 0.34]	[0.11, 0.54]	[-0.11, 0.46]
Control Mean	6.2	3.7	5.3
Observations	251,699	122,017	373,716

Notes: The table replicates the analysis in Table 1 for the outcome of filing a tax return reporting low dollar amounts of earned income. The outcomes considered are indicators for reporting positive earned income less than $1,000 (Panel A), $5,000 (Panel B), and $10,000 (Panel C). Earned income is defined as reported wages and self-employment income. Outcomes take on a value of zero if the individual did not file a tax return. Units are percentage points (0-100). The analysis excludes reporting outcomes for tax year 2023, for which filing data was not available to us at the time of our analysis. Tax returns were excluded from the analysis if they were filed after the end of the calendar year following the tax year covered by the return. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Table A.31: Substitution Elasticities By State

	Work Requirement, Below Phase-Out		No Work Requirement, Above Phase-Out				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	California	Colorado	California	Maryland	New Jersey	Oregon	Vermont
Labor Force Participation							
Effect	0.045 (0.307)	0.247 (0.439)	-0.078 (0.103)	0.243 (0.231)	-0.124 (0.149)	-0.076 (0.334)	-1.725* (0.875)
Control Mean	91.5	93.3	95.9	96.6	97.3	96.0	98.0
Percent Change	0.0	0.3	-0.1	0.3	-0.1	-0.1	-1.8
Return to Work							
Change in Return to Work	1,000	1,450	-1,099	-500	-743	-1,000	-1,000
Control Mean	19,501	20,861	82,009	77,756	117,038	76,763	140,827
Percent Change	8.2	6.3	-1.8	-1.0	-0.7	-1.7	-0.7
Elasticity	0.01 (0.04)	0.04 (0.07)	0.04 (0.06)	-0.26 (0.25)	0.19 (0.22)	0.05 (0.20)	2.48 (1.26)
95% CI	[-0.07, 0.09]	[-0.10, 0.19]	[-0.07, 0.16]	[-0.74, 0.22]	[-0.25, 0.63]	[-0.35, 0.44]	[0.01, 4.94]
Income Range	\$1 - \$25,000	\$1 - \$35,000	$> \$30,000$	$> \$15,000$	$> \$80,000$	$> \$30,000$	$> \$175,000$
Work Requirement	Yes	Yes	No	No	No	No	No

Notes: The table reports estimated extensive-margin substitution elasticities, as described in Section 6. Columns 1 and 2 correspond to state-years for which the applicable credit is conditioned on work; Columns 3-7 correspond to state-years for which the applicable credit is available to non-workers. Column 1 includes California years for 2019 and 2020; Column 3 includes California years for 2022 and 2023. Each column is limited to individuals in the sample for the specified state (reported in 1 and Table 2) who filed a prior-year tax return listing AGI within the specified income range. refers to the estimated percentage point effect of age-eligibility for the applicable credit on labor force participation, following the applicable specification from Table 1 or Table 2. is estimated for mothers in the sample whose children are not age-eligible for the credit for their state. is obtained from dividing the estimated labor force participation effect by the control mean. The change in return to work refers to the simulated effect of age-eligibility for the credit on the net financial gain from working relative to not working, accounting for pre-tax income (measured as prior-year AGI), state and federal income taxes and credits, and SNAP benefits. The simulations are based on prior-year AGI, filing status, and number of dependents. is calculated by dividing the percent change in labor force participation by the percent change in the return to work. The reported standard error and 95% confidence interval for the elasticity are calculated assuming that uncertainty in the labor force participation effect is the only source of uncertainty. For additional details, refer to Appendix C.

Table A.32: Income Elasticities By State

	(1)	(2)	(3)	(4)	(5)
	California	Maryland	New Jersey	Oregon	Vermont
Labor Force Participation					
Effect	-0.094	0.534	0.187	0.508	-0.222
	(0.345)	(1.206)	(0.363)	(1.019)	(0.448)
Control Mean	92.1	92.2	94.1	93.7	96.1
Percent Change	-0.1	0.6	0.2	0.5	-0.2
Non-Working Benefits					
Change in Benefits	1,100	500	752	1,000	1,000
Control Mean	8,207	8,379	8,177	8,823	9,806
Percent Change	14.5	6.4	9.7	12.2	11.0
Elasticity	-0.01	0.09	0.02	0.04	-0.02
	(0.03)	(0.20)	(0.04)	(0.09)	(0.04)
95% CI	[-0.06, 0.04]	[-0.31, 0.49]	[-0.06, 0.10]	[-0.13, 0.22]	[-0.10, 0.06]
Income Range	$1 - $25,000	$1 - $15,000	$1 - $30,000	$1 - $25,000	$1 - $125,000
Work Requirement	No	No	No	No	No

Notes: The table reports estimated extensive-margin income elasticities, as described in Section 6. The analysis is limited to state-years for which the applicable credit was not conditioned on work and to individuals who filed a prior-year tax return reporting AGI within the specified income range. Each column is limited to individuals in the sample for the specified state (reported in 1 and Table 2). refers to the estimated percentage point effect of age-eligibility for the applicable credit on labor force participation, following the applicable specification from Table 1 or Table 2. is estimated for mothers in the sample whose children are not age-eligible for the credit for their state. is obtained from dividing the estimated labor force participation effect by the control mean. The change in Non-Working Benefits refers to the simulated effect of age-eligibility for the credit on the after-tax and transfer income associated with not working, accounting for state and federal refundable tax credits and SNAP benefits. The simulations are based on prior-year AGI, filing status, and number of dependents. is calculated by dividing the percent change in labor force participation by the percent change in the benefits associated with not working. The reported standard error and 95% confidence interval for the elasticity are calculated assuming that uncertainty in the labor force participation effect is the only source of uncertainty. For additional details, refer to Appendix C.

B Work Requirement Earnings Threshold and Labor Participation Response

This Appendix section considers a simple labor supply model to study how taxpayers respond to the $1 work requirement in the California credit as compared to a work requirement tied to a higher earned income threshold.

Individuals choose whether and how much to work. Labor effort is indexed by $l \in \{0\} \cup [1, \infty)$, with $l \geq 1$ denoting work and $l = 0$ denoting not working. Let $u(l)$ refer to the utility in money-metric units to an individual from working $l \in [1, \infty)$ units, accounting for the disutility of work and the utility benefits from earning income, as well as any tax implications or other program benefits other than the program whose work requirement is being considered (e.g., the California credit).

Absent any new program, an individual will choose to work if the utility from working exceeds the utility from not working, for some positive amount of work $l \geq 1$. Thus, the individual will choose to work if and only if the utility associated with the optimal amount of positive work, u_1, exceeds the utility from not working, u_0, where

$$u_1 := \max_l u(l) \text{ s.t. } l \geq 1$$

and u_0 is normalized to zero. The share of the population that works (absent the work requirement under consideration) is therefore given by $Pr(u_1 > 0)$.

Consider a benefit program with a work requirement that provides Y dollars if an individual works at least $k \geq 1$ units, and provides no benefit otherwise. We will refer to this policy as B_k.

Under this policy, if an individual works at least k units, the individual's utility is given by $u_k + Y$, where

$$u_k := \max_l u(l) \text{ s.t. } l \geq k$$

Note that by construction, $k' > k$ implies $u_k \geq u_{k'}$. This follows from the fact that $u_{k'}$ and u_k are defined to equal the maximum of the same function, but the set over which the former is evaluated is a strict subset of the set over which the latter is evaluated: $\{l : l \geq k'\} \subset \{l : l \geq k\}$.

Under policy B_k, an individual's utility from working is the maximum of u_1 and $u_k + Y$. Intuitively, either the individual chooses to work at least k, in which case they obtain utility $u_k + Y$, or they work some positive amount less than k, in which case they would continue to receive u_1 (i.e., the best they could obtain absent the benefit program with the work requirement). Thus, under policy B_k, the individual will work if and only if $u_1 > 0$ or $u_k + Y > 0$. The share of the population that works is thus given by: $Pr(u_1 > 0) + Pr(u_1 \leq 0 \ \& \ u_k \geq -Y)$.

Thus, the effect of introducing policy B_k on the share of the population that works, β_k,

is given by:

$$\begin{aligned}\beta_k &= \underbrace{Pr(u_1 > 0) + Pr(u_1 \leq 0 \ \& \ u_k + Y \geq 0)}_{\text{Share Working Under } B_k} - \underbrace{Pr(u_1 > 0)}_{\text{Share Working Absent } B_k} \\ &= Pr(u_1 \leq 0 \ \& \ u_k + Y \geq 0) \\ &= Pr(u_k + Y \geq 0 \mid u_1 \leq 0)\, Pr(u_1 \leq 0)\end{aligned}$$

Finally, note that for any $k' > k$, we have:

$$\begin{aligned}\beta_{k'} &= Pr(u_{k'} + Y \geq 0 \mid u_1 \leq 0)\, Pr(u_1 \leq 0) \\ &\leq Pr(u_k + Y \geq 0 \mid u_1 \leq 0)\, Pr(u_1 \leq 0) \\ &= \beta_k\end{aligned}$$

where the inequality follows from the fact (discussed above) that for each individual $k' > k$ implies $u_k \geq u_{k'}$.

C Labor Supply Elasticity Calculations

This Appendix provides additional detail on the elasticity calculations based on our estimates of the effect of child tax credit eligibility on labor force participation.

The outcome of interest is mean labor force participation, denoted by Y. Let $Y(1)$ denote mean labor force participation among those whose children are age-eligible for the applicable credit, and let $Y(0)$ denote mean labor force participation among those who children are not age-eligible for the applicable credit.

The extensive-margin substitution elasticity we consider is defined as:

$$\frac{\%\Delta \text{ labor force participation}}{\%\Delta \text{ return to work}}. \tag{1}$$

The numerator of (1) reflects the percent change in the fraction of the population that works based upon age-eligibility for the credit, defined as:

$$\%\Delta \text{ labor force participation} = \frac{Y(1) - Y(0)}{Y(0)} \tag{2}$$

The denominator of (1) reflects the percent change in the return-to-work, RTW, accounting for taxes and transfers:

$$\%\Delta \text{ return to work} = \frac{RTW(1) - RTW(0)}{RTW(0)} \tag{3}$$

where $RTW(1)$ and $RTW(0)$ denote the mean after-tax-and-transfer return-to-work with and without age-eligibility for the credit. In turn, the after-tax-and-transfer return-to-work is defined as the difference in income one would receive from working compared to not working, accounting for pre-tax income, federal and state income taxes (including credits), and SNAP benefits.

We calculate the after-tax-and-transfer return to work for various income levels ranging from \$100 to \$200,000 (in \$100 increments) for various household types (marital status, number of children and state of residence) and calculate federal and state taxes (including refundable tax credits) for each household type and earnings level using NBER's TAXSIM.[54] In some cases we manually model state child tax credits that have not yet been fully incorporated into TAXSIM. Because California's work requirement induced a very large return to work in percentage terms for individuals with near-zero income, we group individuals with income below \$100 into the \$100 bin to limit the influence of these outliers on the mean return-to-work for our sample. We calculate SNAP benefits based on household (assumed to be the tax unit) income and size, using the benefit formula for the federal SNAP program.[55]

[54]Each household is assigned a single RTW value, and married couples' RTW is modeled jointly. For simulated married households, we do not separately model each adult's RTW given their spouse's earnings levels.

[55]This calculation uses a benefits reduction rate of 30%.

We assume that households with no earned income would receive the maximum SNAP benefits available given their household size. Households that do not work are technically eligible for other programs such as cash welfare. Because take up of those benefits tend to hover in the 20-30% range, we disregard the value of cash welfare for this exercise.

The extensive-margin income elasticity we consider is defined as

$$\frac{\%\Delta \text{ labor force participation}}{\%\Delta \text{ benefits from not working}}. \tag{4}$$

As in (2), the numerator of (4) reflects the percent change in the fraction of the population that works based on age-eligibility for the credit.

The denominator of (4) reflects the percent change in the after-tax and transfer benefits one would obtain from not working based on age-eligibility for the credit:

$$\%\Delta \text{ benefits from not working} = \frac{BNW(1) - BNW(0)}{BNW(0)} \tag{5}$$

where $BNW(1)$ and $BNW(0)$ denote the mean after-tax-and-transfer benefits one would receive from not working with and without age-eligibility for the credit. As above, the benefits from not working account for federal and state refundable tax credits as well as SNAP benefits.

We calculate population-weighted average RTW estimates from our administrative data. We assign each adult in our sample the RTW estimate from our simulation given their marital status, state of residence, number of children, and prior year AGI.